

Data Engineering on Google Cloud

— Day 2 (of 2)

Iowa Department of Management, Department of IT · October 6, 2026 · Student slide pack

How to use this pack

- Slides follow the class agenda — each module section opens with a divider page showing its time slot.
- Hands-on activities (instructor-led demos you follow along):
<https://roitraining.github.io/gcp-demos/bigquery.html>
- Labs are in the Google Skills classroom — open the classroom, launch the lab listed for each block.

Agenda — Tuesday, October 6, 2026 (times CT)

Time	Block
09:00–09:15	Recap quiz + spillover
09:15–09:45	Module 7 — Intro to Modern Data Engineering (lake vs warehouse vs lakehouse)
09:45–10:30	Module 8 — Building a Data Lakehouse
10:30–10:45	Break
10:45–11:15	Lab — Federated Query with BigQuery
11:15–11:35	External tables — Bronze made tangible (activity \$30)
11:35–12:00	Module 9 — Modernizing Data Warehouses with BigQuery & BigLake
12:00–01:00	Lunch
01:00–01:45	Lab — Querying External Data and Iceberg Tables
01:45–02:00	Iceberg deep-dive demo
02:00–02:35	Module 10 — Governance + the Medallion (Bronze/Silver/Gold)
02:35–02:50	Break
02:50–03:20	Module 11 — BQML + the Power BI last mile (compressed)
03:20–03:50	Capstone workshop — map one of your real extract jobs
03:50–04:10	Course summary — services come together + key points
04:10–04:30	Flex / Q&A / evaluations

MODULE 7

Introduction to Modern Data Engineering on Google Cloud

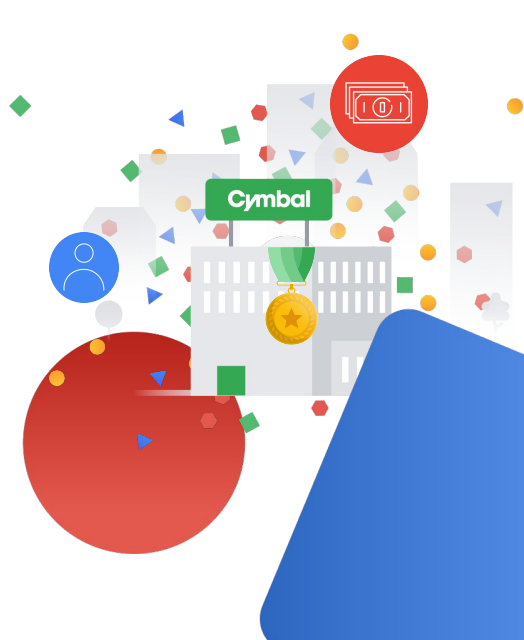
· 09:15–09:45 ·

slides 2–21 + quiz 24–32

source deck: M1_Introduction_to_modern_data_engineering_on_Google_Cloud.pdf

For years, organizations have relied on two primary methods for large-scale data storage: **data lakes and data warehouses.**

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How are organizations, such as Cymbal (an e-commerce company), **managing extensive datasets** with data lakes and data warehouses?

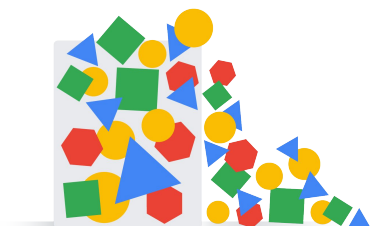
To illustrate how data lakes and data warehouses are used, let's examine the challenges faced by Cymbal, an e-commerce company. Cymbal has a vast product portfolio. This means they have to work with a lot of data behind the scenes on a daily basis.

Data lakes and data warehouses serve different purposes and are optimized for different kinds of data and workloads.

To understand them, let's consider how they may be utilized in Cymbal's data environment.

Data lakes

Think of a data lake as a vast reservoir of data. It stores enormous amounts of raw data in its native format.



01 Structured data

02 Semistructured data

03 Unstructured data

Think of a data lake as a vast reservoir of data. It stores enormous amounts of raw data in its native format. You can pour any kind of data into it; structured, semistructured, and unstructured.

For Cymbal, this means storing:

1. Structured data, such as transaction tables from its sales database.
2. Semistructured data, such as JSON logs from its web servers.
3. Unstructured data, such as customer-submitted product images, videos, and text reviews.

Data lakes

01

Data lakes operate on the principle of "store everything now, figure out how to use it later."

02

This approach applies a structure to data upon reading (schema-on-read), not during storage.

03

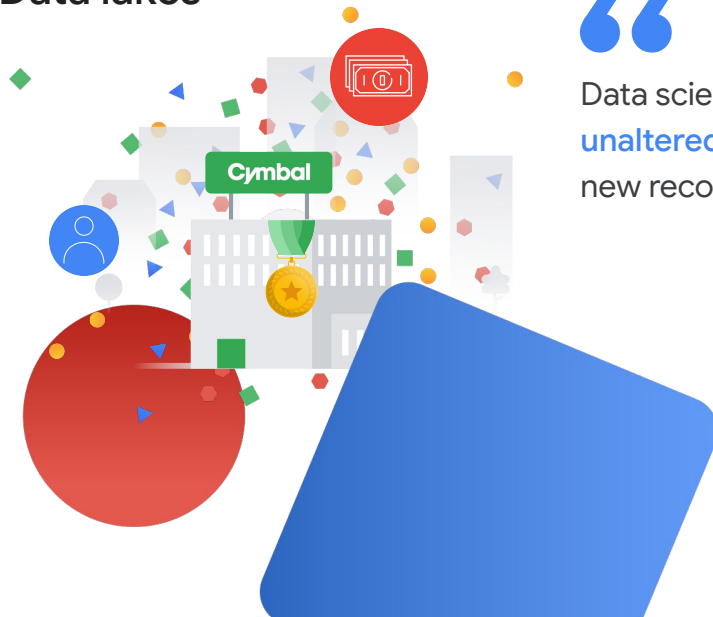
Data lakes are well suited for data exploration, machine learning (ML), and big data processing.

Data lakes are created on the premise of "store everything now, figure out how to use it later."

With this approach, you apply a structure to the data when you read it (schema-on-read), not when you store it.

This makes data lakes ideal for data exploration, machine learning (ML) and big data processing.

Data lakes



“

Data scientists need access to original, **unaltered clickstream data** to build a new recommendation engine.



Cymbal's data scientists, for example, need access to original, unaltered clickstream data to build a new recommendation engine.

Data lakes accomplish this by storing large quantities of unaltered data from various sources, which allows Cymbal's data scientists to find the best elements to create a robust and accurate model.

Data lakes: Advantages

- 01 **Flexibility:** Stores all data types.
- 02 **Agility:** Fast to ingest data.
- 03 **Scalability:** Can grow to exabyte scale.
- 04 **Cost-effectiveness:** Uses low-cost object storage.
- 05 **Support for advanced analytics:** Ideal for AI/ML model training.

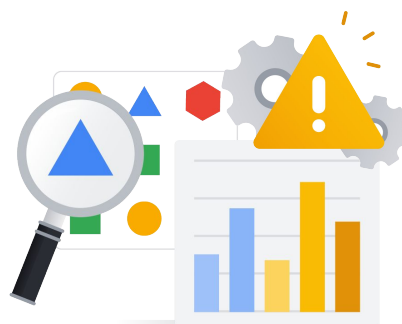


Data lakes offer the following key advantages:

1. Unparalleled flexibility with all data types.
2. Rapid data ingestion for timely insights.
3. Impressive scalability to exabyte scale.
4. Cost-effectiveness with low-cost object storage.
5. Advanced analytics support for AI/ML model training.

Data lakes: Disadvantages

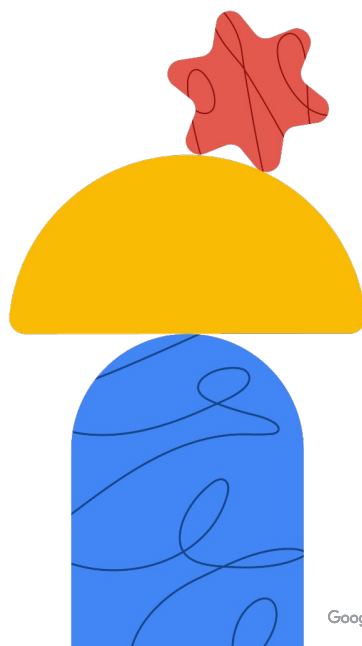
- 01 **Risk of becoming a "data swamp":** Can become a disorganized collection of unusable data.
- 02 **Management complexity:** Requires significant overhead to maintain.
- 03 **Time-consuming analysis:** Data needs to be cleaned and wrangled before it is usable.
- 04 **Security risks:** Raw data formats can increase security and compliance risks.



The disadvantages of data lakes, on the other hand, include the following:

1. Without proper governance, data lakes can quickly become disorganized and unusable, essentially turning into a "data swamp."
2. They require significant ongoing maintenance and management, which can be a substantial overhead for organizations.
3. Before data can be analyzed, it often needs extensive cleaning and wrangling, adding time to the insights generation process.
4. Raw data formats often stored in data lakes can increase security and compliance risks if not properly managed.

A **data warehouse** is
like a highly organized
library.



Google Cloud

In contrast, a data warehouse is like a highly organized library.

Let's learn more.

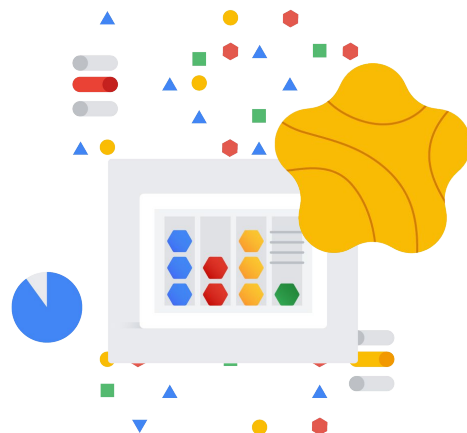
Data warehouses

01

Data is cleaned, transformed, and structured before storage.

02

This process, schema-on-write, optimizes data for analysis and business intelligence (BI).



In a data warehouse, data is cleaned, transformed, and structured before it's stored.

This process, known as schema-on-write, ensures the data is optimized for analysis and business intelligence (BI).

Data warehouses

Structure and quality first

The intended use of the data must be determined before the data warehouse is designed.

Cymbal example

- Key metrics, such as customer lifetime value and daily sales figures, are already defined.
- The data warehouse is then built to provide fast and accurate reports on these specific metrics.



The premise for data warehouses is "structure and quality first." The intended use of the data must be determined before the data warehouse is designed.

For Cymbal, this means the business has already defined key metrics, such as customer lifetime value and daily sales figures. The data warehouse is then built to provide fast and accurate reports on these specific metrics.

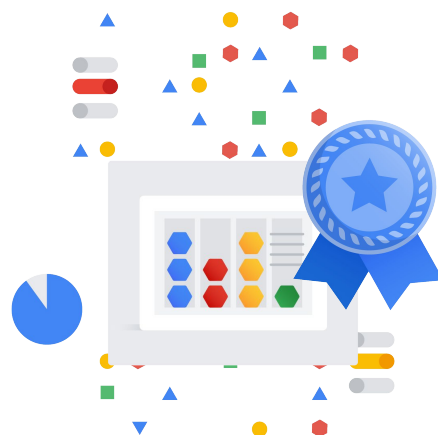
Data warehouses are perfect for generating business reports and dashboards where **speed and accuracy are critical.**

Data warehouses are perfect for generating business reports and dashboards where speed and accuracy are critical.

Let's examine their key advantages and disadvantages for a comprehensive understanding.

Data warehouses: Advantages

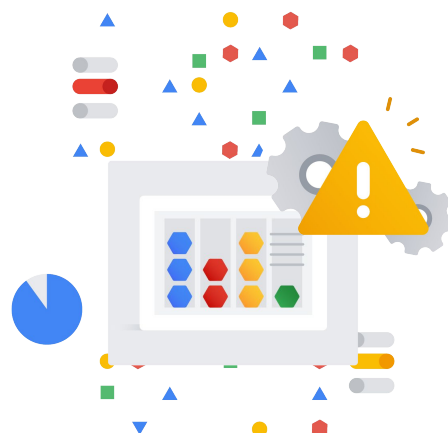
- 01 **Speed:** Optimized for high-performance queries.
- 02 **High-quality data:** Provides consistent and reliable information.
- 03 **Business focused:** Directly answers common business questions.
- 04 **Historical intelligence:** Enables deep analysis of trends over time.



1. First, data warehouses provide speed, as they are optimized for high-performance queries.
2. Second, you get high-quality data, ensuring consistent and reliable information.
3. Third, data warehouses are business focused, directly answering common business questions.
4. Finally, they enable historical intelligence, allowing for deep analysis of trends over time.

Data warehouses: Disadvantages

- 01 **Inflexibility:** Cannot easily accommodate new data types or unstructured data.
- 02 **Cost:** Often expensive to build and maintain.
- 03 **Limited data types:** Primarily designed for structured data only.
- 04 **Long development time:** Can take a long time to design and implement.



When discussing the disadvantages, we need to highlight several key points:

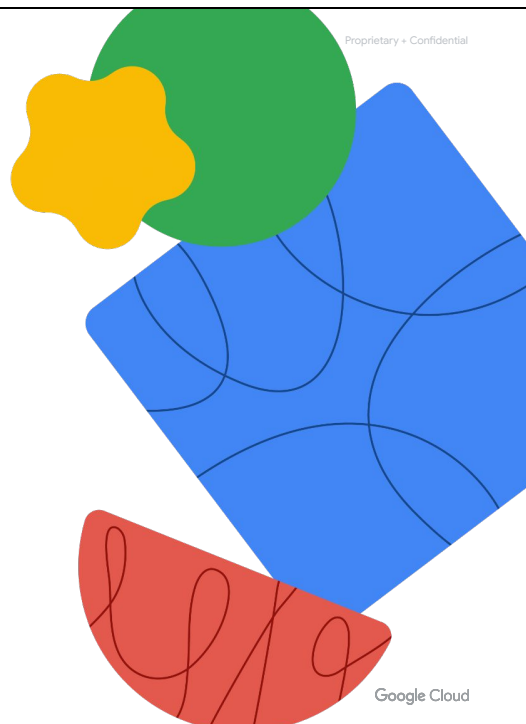
1. **Inflexibility:** Data warehouses struggle to accommodate new data types or unstructured data due to their rigid structure.
2. **Cost:** They are often expensive to build and maintain, requiring significant investment.
3. **Limited data types:** Primarily designed for structured data, they may not suit organizations with diverse, unstructured data.
4. **Long development time:** Designing and implementing a data warehouses can take considerable time, delaying benefits.

Neither data lakes or data warehouses excel at supporting the **full range of an organization's data needs.**

Relying on a single design approach can be challenging, as neither excels at supporting the full range of an organization's data needs.

The modern approach: Data lakehouse

Combines the flexibility and low cost of a data lake with the speed and accuracy of a data warehouse.



Google Cloud

Due to their complementary nature, an ideal approach combines the flexibility and low cost of a data lake with the speed and accuracy of a data warehouse.

This hybrid approach has created the concept of a data lakehouse.

Data lakehouse

01

Unified platform

Combines low-cost storage of a data lake with data warehouse management and query performance.

02

Versatile workloads

Supports traditional BI, modern data science, and AI without data movement or duplication.

03

Optimized architecture

Implements metadata and governance on open-format files in low-cost object storage (such as Google Cloud Storage) for a hybrid approach.

Google Cloud

1. A data lakehouse architecture combines the low-cost storage of a data lake with the management features and query performance of a data warehouse.
2. The goal is to create a single, unified platform that can support traditional BI, modern data science, and AI workloads without moving or duplicating data.
3. A data lakehouse achieves this by implementing a metadata and governance layer on top of open-format files stored in low-cost object storage, such as Google Cloud Storage. This gives you the best of both worlds.

Data lakehouse

Data silo challenges

Cymbal's sales data was in a data warehouse and its customer review data was in a data lake. The two could not be analyzed together easily.

Integrated data analysis

With a data lakehouse, Cymbal can now run a single query to correlate customer sentiment from text reviews with sales trends from its transaction database.

For Cymbal, a data lakehouse solves a key challenge: breaking down data silos.

Previously, its sales data was in a data warehouse and its customer review data was in a data lake. The two could not be analyzed together easily.

With a data lakehouse, Cymbal can now run a single query to correlate customer sentiment from text reviews with sales trends from its transaction database.

Data lakehouse: Key features

01 Support for most data formats

02 Flexible schema-on-read or schema-on-write

03 Access for all types of data users

04 Cost flexibility based on needs

05 Unified data governance

06 ACID transaction support

The key features of a Google Cloud data lakehouse include:

1. Support for most data formats
2. Flexible schema-on-read or schema-on-write
3. Access for all types of data users
4. Cost flexibility based on needs
5. Unified data governance
6. ACID transaction support



Lakehouse technology

- Google Cloud uses Lakehouse to implement data lakehouses.
- Lakehouse extends Google Cloud's governance and query features to data in Google Cloud or other major clouds.
- Manage and query data in its original location using open formats, such as Parquet, ORC, and Avro.
- Utilize BigQuery's powerful query engine.

Google Cloud

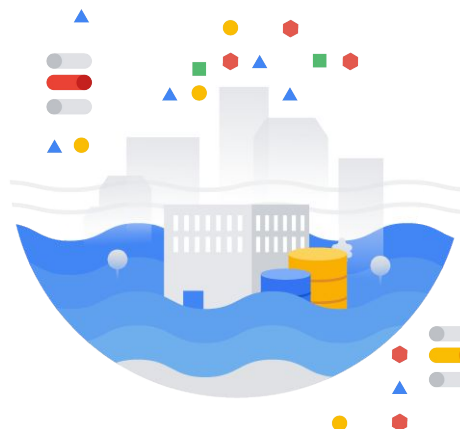
Google Cloud implements data lakehouses through Lakehouse for Apache Iceberg (Lakehouse) technology.

Lakehouse allows you to apply Google Cloud's governance and query capabilities to data stored in Google Cloud or even in other major clouds.

You can manage and query your data where it lives using open formats, such as Parquet, ORC, and Avro, while benefiting from BigQuery's powerful and flexible query engine.

Data lakehouse benefits

- Unified data storage and access
- Reduced data redundancy
- Unified governance
- Broken-down data silos
- Improved flexibility and scalability

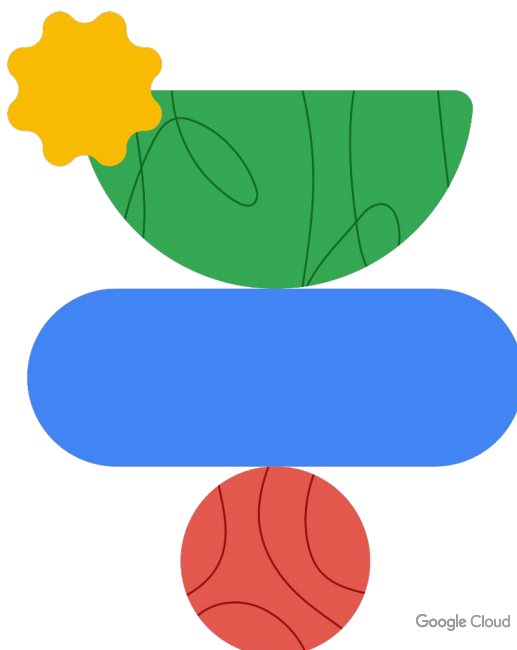


Data lakehouses solve many challenges by unifying data storage and access.

The benefits include:

- Reduced data redundancy
- Unified governance
- Broken-down data silos
- Improved flexibility and scalability

Quiz



Quiz | Question 01

Question

Which of the following best describes the primary characteristic of a data lake?

- A. Data is cleaned, transformed, and structured before storage, optimized for BI.
- B. It is a highly organized library designed for fast and accurate reports based on predefined metrics.
- C. It stores enormous amounts of raw data in its native format with schema-on-read, ideal for exploration and ML.
- D. It primarily handles structured data and is optimized for high-performance queries on consistent information.

Quiz | Question 01

Answer

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- C. It stores enormous amounts of raw data in its native format with schema-on-read, ideal for exploration and ML.
- D. It primarily handles structured data and is optimized for high-performance queries on consistent information.



Option A is incorrect because it describes a data warehouse, which uses a "schema-on-write" approach where data is structured before being stored.

Option B is incorrect as it refers to a data warehouse, which is compared to a "highly organized library" optimized for specific business reports.

* Option C is correct. The lesson defines a data lake as a "vast reservoir" that stores raw data in its native format using a "schema-on-read" approach ideal for data exploration and ML.

Option D is incorrect because it describes a data warehouse, which is designed for structured data and high-performance queries. Data lakes, in contrast, are built to handle structured, semistructured, and unstructured data.

Quiz | Question 02

Question

What is a key disadvantage of a traditional data warehouse?

- A. It risks becoming a "data swamp" without proper governance.
- B. It is inflexible and cannot easily accommodate new data types or unstructured data.
- C. It primarily uses low-cost object storage, leading to security risks with raw data.
- D. It is ideal for AI/ML model training due to its support for advanced analytics.

Quiz | Question 02

Answer

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Option A is incorrect because the risk of becoming a "data swamp" is a disadvantage of data lakes, not data warehouses.

* Option B is correct. The lesson explicitly states inflexibility as a disadvantage of data warehouses, stating they "cannot easily accommodate new data types or unstructured data."

Option C is incorrect because these are characteristics of a data lake, which uses low-cost storage and can have security risks due to raw data formats.

Option D is incorrect because the lesson states that data lakes, not data warehouses, are ideal for AI/ML model training.

Quiz | Question 03

Question

How does a data lakehouse architecture combine the best features of data lakes and data warehouses?

- A. By enforcing a strict schema-on-write approach for all data types to ensure high data quality.
- B. By creating separate, specialized platforms for BI and AI workloads to maximize performance.
- C. By implementing a metadata and governance layer on top of open-format files in low-cost object storage.
- D. By always requiring data to be moved and duplicated between a data lake and a data warehouse.

Quiz | Question 03

Answer

How does a data lakehouse architecture combine the best features of data lakes and data warehouses?

- A. By enforcing a strict schema-on-write approach for all data types to ensure high data quality.
- B. By creating separate, specialized platforms for BI and AI workloads to maximize performance.
- C. By implementing a metadata and governance layer on top of open-format files in low-cost object storage.
- D. By always requiring data to be moved and duplicated between a data lake and a data warehouse.



Option A is incorrect because a key feature of a data lakehouse is its flexibility; it supports both schema-on-read (like a data lake) and schema-on-write (like a data warehouse), rather than enforcing a single, strict approach.

Option B is incorrect because the goal of a data lakehouse is to create a single, unified platform to support various workloads, thereby breaking down the data silos that separate platforms create.

* Option C is correct because the lesson explicitly states that a data lakehouse achieves its hybrid functionality by implementing a "metadata and governance layer on top of open-format files stored in low-cost object storage, such as Google Cloud Storage." This approach provides warehouse-like management for data stored in a data lake-like environment.

Option D is incorrect because a primary advantage of the data lakehouse is that it supports BI, data science, and AI workloads "without moving or duplicating data," which reduces redundancy.

Quiz | Question 04

Question

Cymbal, an e-commerce company, needs a solution to support traditional BI, modern data science, and AI workloads on a single, governed copy of its data while also breaking down data silos between sales and customer review data. Which architecture is best suited for this need?

- A. A data warehouse, such as BigQuery, for high-speed, interactive BI on structured data.
- B. A data lake, using Cloud Storage, for initial, low-cost storage of massive volumes of raw data for AI/ML research.
- C. A data lakehouse architecture with BigQuery and Lakehouse.
- D. Separate data lakes and data warehouses, with manual data integration as needed.

Quiz | Question 04

Question

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- C. A data lakehouse architecture with BigQuery and Lakehouse.
- D. Separate data lakes and data warehouses, with manual data integration as needed.



Google Cloud

Option A is incorrect because a data warehouse alone is not ideal for the AI and data science workloads that require access to raw, unstructured data, such as customer reviews. It would only satisfy the company's BI needs.

Option B is incorrect because a data lake by itself is not optimized for high-performance queries and reporting required for traditional BI. It would only serve the AI and research needs effectively.

* Option C is the correct choice. The lesson explicitly recommends this architecture "when you need to do it all: support BI, AI, and data science on a single, governed copy of your data." It also directly addresses Cymbal's challenge by breaking down data silos, allowing the company to run a single query correlating customer reviews from the data lake with sales trends from the data warehouse.

Option D is incorrect because this approach perpetuates the very data silos that a data lakehouse is designed to break down. Maintaining separate systems is inefficient and makes it difficult to get a holistic view of the business.

MODULE 8

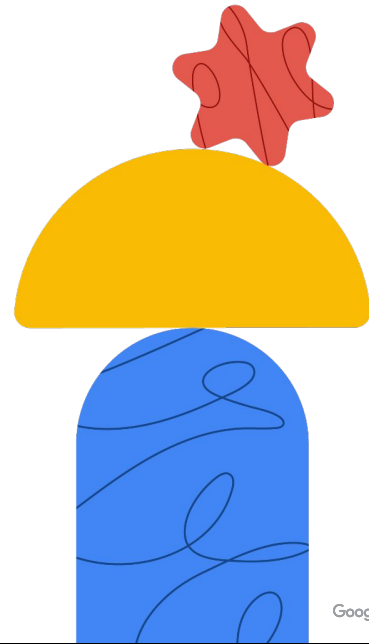
Building a Data Lakehouse with Cloud Storage, Open Formats, and BigQuery

· 09:45–10:30 ·

slides 5–8, 13–17, 18–21, 27–28, 45 (lab intro)

source deck: [M2_Building_a_Data_Lakehouse_with_Cloud_Storage,_Open_Formats,_and_BigQuery.pdf](#)

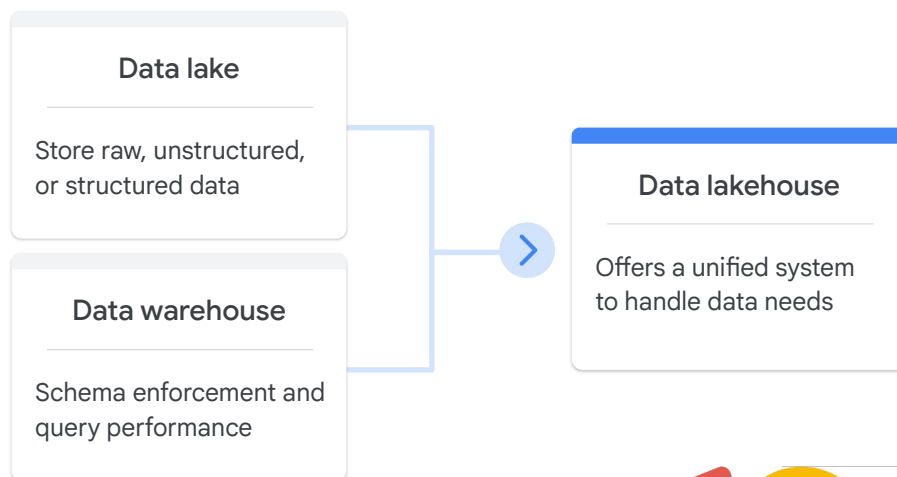
Data storage for data lakehouses on Google Cloud



Google Cloud

So far, you've learned that a data lakehouse combines the key features of data lakes and data warehouses.

Now let's explore in a little more detail what that means for data storage in a Google Cloud data lakehouse.



A data lakehouse offers the flexibility and scalability of a data lake for storing raw, unstructured, or structured data, with the schema enforcement and query performance of a data warehouse.

This hybrid approach allows organizations, such as Cymbal, to handle all their data needs in one unified system.

Google Cloud provides a powerful set of tools to build this unified architecture.

Cloud Storage

A massively scalable, highly durable, and cost-effective object storage service.



The primary storage for your data lakehouse is Cloud Storage. Cloud Storage is a massively scalable, highly durable, and cost-effective object storage service.

Cymbal can use it to store almost any type of data file, regardless of its size or format. This includes unstructured data, such as user comments, and even files, such as images.

Other storage options

If your strategy is multicloud, you can keep your data in another major provider's cloud storage.

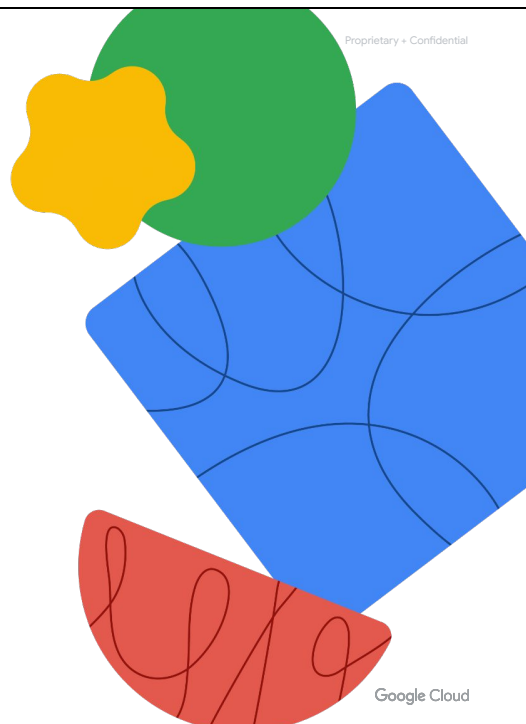


However, you are not limited to using Google Cloud Storage.

If your strategy is multicloud, you can keep your data in another major provider's cloud storage.

Introduction to Apache Iceberg open table format

A data lakehouse needs structure and performance. Open table formats, such as Apache Iceberg provide this.

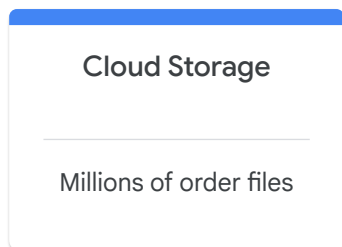


Google Cloud

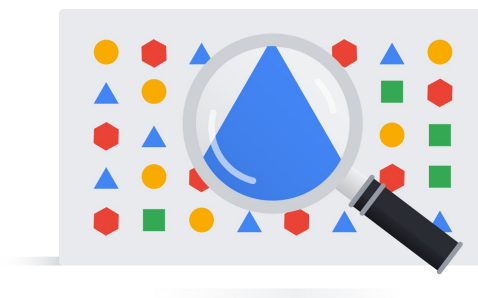
Although Cloud Storage is excellent for storing raw files, a data lakehouse needs a way to bring structure and performance to this data.

This is where open table formats come into play, and Apache Iceberg, is a leading example.

Efficient querying with Apache Iceberg (1 of 2)



Without Apache
Iceberg: Query scans
every file

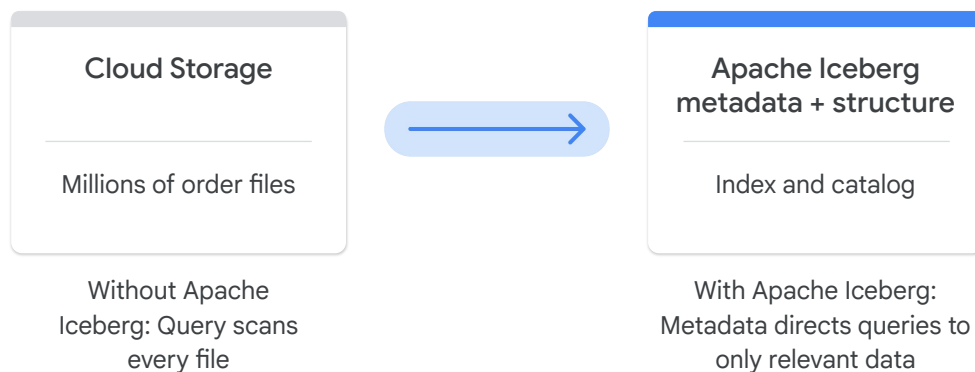


Cymbal has millions of customer orders stored as files in Cloud Storage.

If they want to query these orders efficiently, for example, to find all orders placed in the last week by customers in a specific region, traditional file-based queries would be slow.

They would have to scan every file to find the relevant information.

Efficient querying with Apache Iceberg (2 of 2)



Apache Iceberg solves this by adding a layer of metadata and structure on top of the files in Cloud Storage.

It does not move your data; it organizes and manages it, acting as an index and catalog for your data files.

What does Apache Iceberg provide?

01

Schema evolution

Allows schema changes (such as adding and renaming columns) without impacting existing data or applications.

02

Hidden partitioning

Automatically manages how data is partitioned for efficient querying.

03

Time travel

Enables querying past data versions for auditing and debugging, allowing users to reproduce exact data states from previous points in time.

04

Atomic transactions

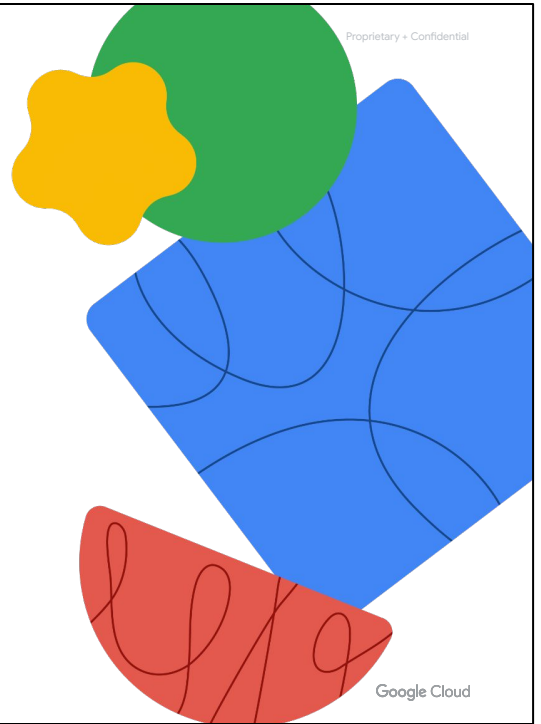
Enable reliable, concurrent data operations, preventing corruption when multiple processes read/write to the same tables.

Apache Iceberg provides the following:

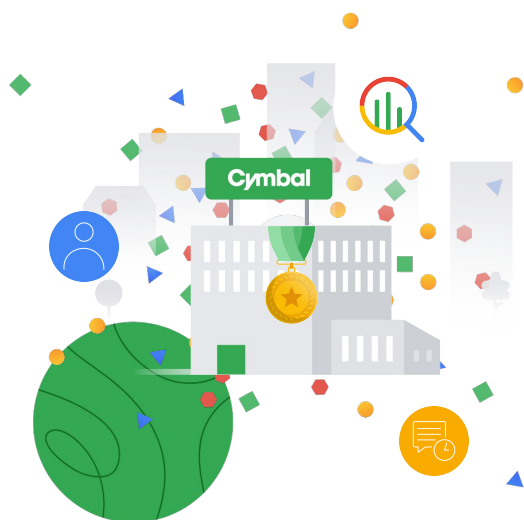
1. Schema evolution: It allows schema changes (such as adding and renaming columns) without impacting existing data or applications.
2. Hidden partitioning: Apache Iceberg automatically manages how data is partitioned for efficient querying. For Cymbal's analysts, this means not having to worry about the underlying file organization.
3. Time travel: This allows Cymbal to query past versions of their data. If a report was run last month, they can reproduce the exact data state from that time, which is invaluable for auditing and debugging.
4. Atomic transactions: Apache Iceberg enables reliable, concurrent operations on the data, ensuring that multiple processes can read and write to the same tables without data corruption.

Using Apache Iceberg with Cloud Storage

Treat a collection of data files as performant, queryable tables, bringing the benefits of a data warehouse to a flexible data lake.



Using Apache Iceberg with Cloud Storage, Cymbal can treat their collection of data files as performant, queryable tables, bringing the benefits of a data warehouse to their flexible data lake.



BigQuery as the central processing engine

Whereas Cloud Storage and Apache Iceberg provide a flexible, open-standard foundation for storing vast amounts of raw and structured data, BigQuery is the high-performance engine that activates it.

BigQuery serves as the central analytics and query engine for Cymbal's data lakehouse.

Whereas Cloud Storage and Apache Iceberg provide a flexible, open-standard foundation for storing vast amounts of raw and structured data, BigQuery is the high-performance engine that activates it.

BigQuery

Seamless data querying with BigQuery and Lakehouse

By creating Lakehouse tables, Cymbal can use BigQuery's familiar SQL interface to directly and securely query the Apache Iceberg-formatted data in their Cloud Storage data lake.

Eliminating data duplication and ETL processes

No need to duplicate data or perform costly extract, transform, load (ETL) processes to run analytics.

Enhanced analytics performance for data lake

Analysts get the performance and features of a premier data warehouse while querying data directly in their open-format data lake.

By creating Lakehouse tables, Cymbal can use BigQuery's familiar SQL interface to directly and securely query the Iceberg-formatted data in their Cloud Storage data lake.

This means they do not have to duplicate data or perform costly extract, transform, load (ETL) processes to run analytics.

Analysts get the performance and features of a premier data warehouse while querying data directly in their open-format data lake.

BigQuery

Native storage for “hot” datasets

BigQuery offers optimized, managed storage for frequently accessed or "hot" datasets. This is ideal for data requiring the fastest query performance, such as key marketing performance dashboards.

Unified data analysis platform

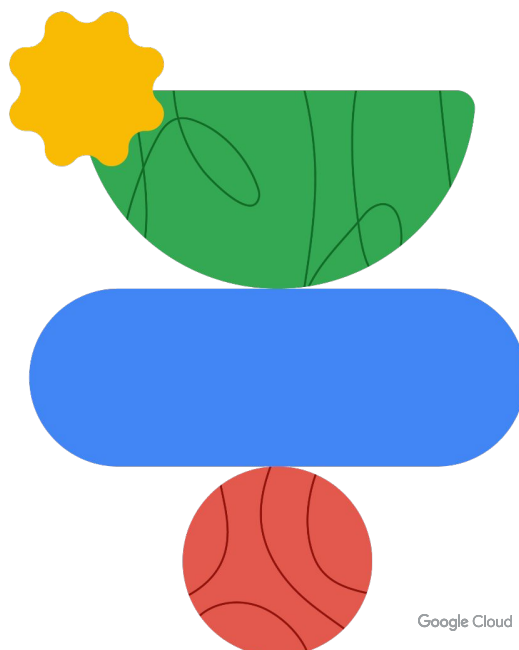
Cymbal can leverage a unified platform to analyze data located in their data lake, BigQuery's native storage, or a combination of both, all accessible through a single interface.

Furthermore, BigQuery also provides its own optimized, managed storage.

Cymbal can use this native storage for their most frequently accessed or "hot" datasets that require the fastest query performance, such as key marketing performance dashboards.

This gives them a unified platform to analyze data in their data lake, in BigQuery's native storage, or both, all through a single interface.

Combining operational data in AlloyDB



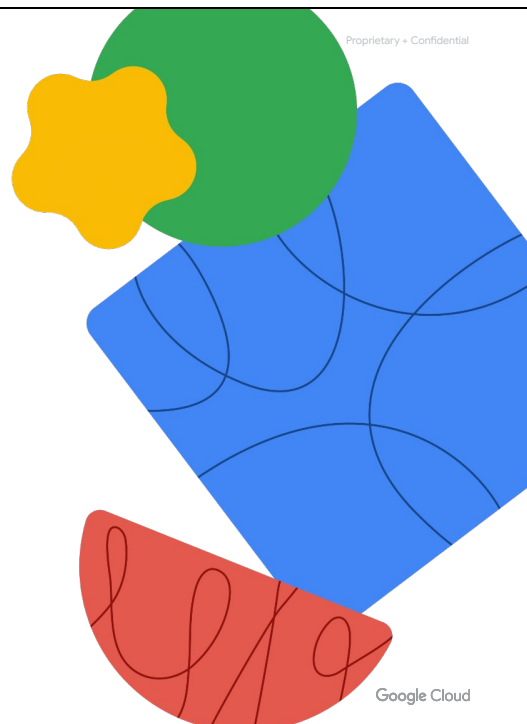
Google Cloud

Whereas Cloud Storage and open table formats, such as Apache Iceberg, are ideal for large-scale analytical data, operational data that powers Cymbal's daily operations require a different solution.

This is where AlloyDB for PostgreSQL is an excellent choice.

Federated queries: Combining operational and analytical data in a single query

Query data in external systems, such as an operational AlloyDB database, in real time without moving or copying it into BigQuery.



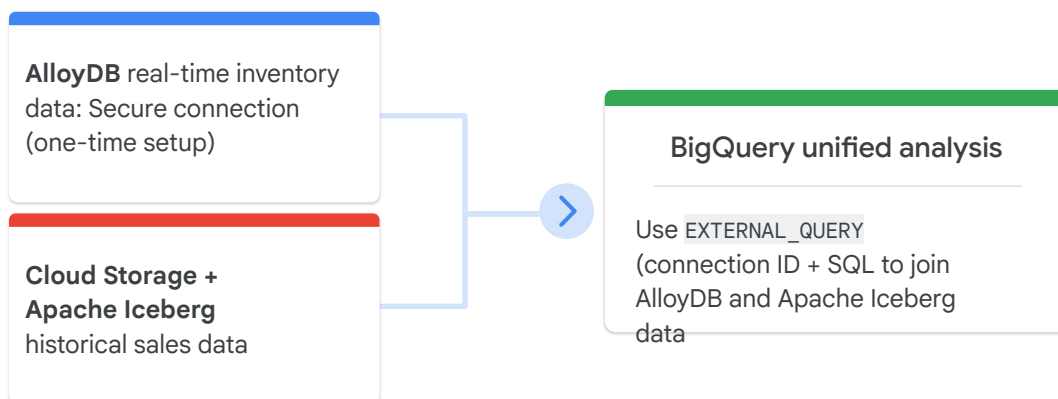
Google Cloud

BigQuery's capabilities extend beyond the data lakehouse through federated queries.

This feature allows Cymbal to query data in external systems, such as their operational AlloyDB database, in real time without moving or copying it into BigQuery.

This creates a bridge between their live operational data and their historical analytical data by allowing queries that combine both operational and analytical data.

Combining data



Here is how Cymbal would combine real-time inventory data from AlloyDB with historical sales data from their Apache Iceberg tables:

1. First, a data administrator creates a secure connection resource in BigQuery. This connection contains the credentials and configuration details to allow BigQuery to communicate with Cymbal's AlloyDB instance. This is a one-time setup.
2. To access the AlloyDB data, an analyst uses the `EXTERNAL_QUERY` SQL function within a BigQuery query. This function takes two arguments: the connection ID and the query to be executed on the AlloyDB database.

Lab: Federated query with BigQuery and AlloyDB

Overview

Explore how federated queries allows you to gain real-time insights by directly querying an operational database from within BigQuery.



What you'll do

- Connect BigQuery to an AlloyDB for PostgreSQL database.
- Join historical analytical data with live transactional data.

In this lab, you will explore the power of federated queries to gain real-time insights by directly querying an operational database from within BigQuery.

What you'll do:

- Connect BigQuery to an AlloyDB for PostgreSQL database.
- Join historical analytical data with live transactional data.

MODULE 9

Modernizing Data Warehouses with BigQuery and Lakehouse

· 11:35–12:00 ·

instructor slides 4–16, 20–28, 30–35, 36–42, 62 (lab intro)

(student pages remapped +4 after p25 — student deck has 4 extra demo-transcript pages)

source deck: M3_Modernizing_data_warehouses_with_BigQuery_and_Lakehouse.pdf

Traditional data warehouse challenges (1 of 2)

Cymbal's data architects face significant challenges with traditional data warehouses:

01

Extensive time spent on capacity planning for events, such as the holiday shopping season.

03

Manual provisioning of hardware, software installation, and patching.

02

Difficulty handling unexpected traffic spikes.

04

Rebalancing data manually as it grows.

Cymbal's data architects would have to spend a significant amount of time on capacity planning.

How many servers do we need for the upcoming holiday shopping season? How do we handle unexpected spikes in traffic?

They would be responsible for provisioning hardware, installing and patching software, and manually rebalancing data as it grows.

Traditional data warehouse challenges (2 of 2)

Operational overhead is **expensive** and **slows down** the ability of the analytics teams to get **timely insights**.

01

Extensive time spent on capacity planning for events, such as the holiday shopping season.

03

Manual provisioning of hardware, software installation, and patching.

02

Difficulty handling unexpected traffic spikes.

04

Rebalancing data manually as it grows.

This operational overhead is not only expensive but also slows down the ability of the analytics teams to get timely insights.

This is the exact problem that BigQuery was designed to solve.

BigQuery

Enterprise-grade, fully managed,
serverless data warehouse.



BigQuery is Google's enterprise-grade, fully managed, serverless data warehouse.

Let's break down what that really means.

BigQuery fundamentals

Fully managed

The infrastructure (such as the hardware, the networking, and the low-level software) is all handled by Google. Your team doesn't need to worry about patches, updates, or hardware failures.

Serverless

Serverless takes things a step further. You don't have to provision or manage any servers at all. You simply load your data and start querying. BigQuery automatically allocates the necessary resources to run your queries and scales them up or down based on the complexity of your request.

The infrastructure (such as the hardware, the networking, and the low-level software) is all handled by Google. Your team doesn't need to worry about patches, updates, or hardware failures.

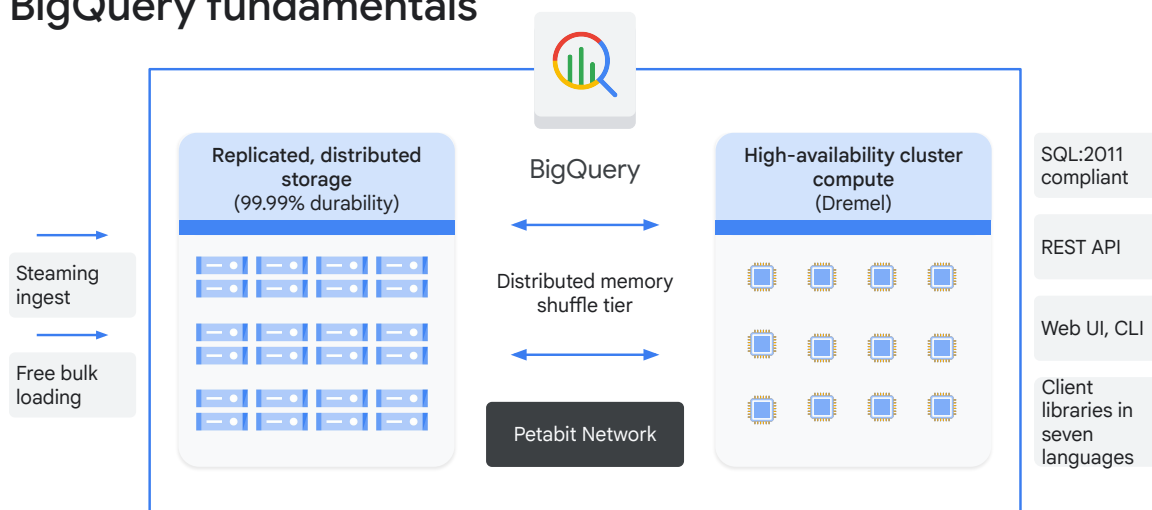
Serverless takes things a step further. You don't have to provision or manage any servers at all.

You simply load your data and start querying. BigQuery automatically allocates the necessary resources to run your queries and scales them up or down based on the complexity of your request.

The magic behind BigQuery is its architecture. BigQuery **separates storage from compute.**

The magic behind BigQuery is its architecture. BigQuery separates storage from compute.

BigQuery fundamentals



Think of it like a library. The books are the data stored reliably and inexpensively in Google's distributed file system.

When you want to find specific information, the librarians are the compute resources.

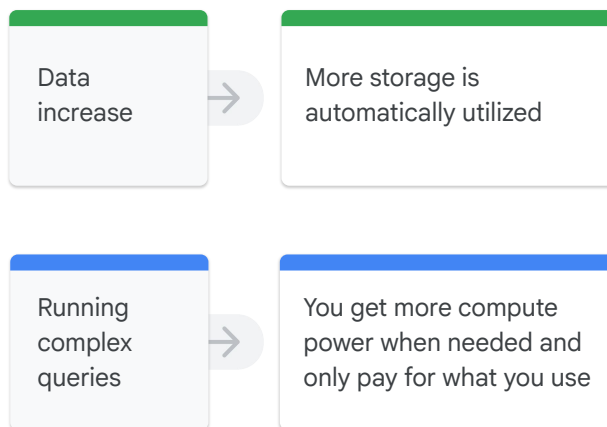
BigQuery can call upon thousands of librarians (or compute workers) simultaneously to scan the entire library (your data) very quickly.

This distributed processing engine, called Dremel, is what makes your queries run so fast.

BigQuery fundamentals

Because storage and compute are separate, they can scale independently.

This separation is a game changer for managing costs and ensuring performance.



Because storage and compute are separate, they can scale independently. If Cymbal's data grows, more storage is automatically utilized.

If they need to run more complex queries, they can use more compute power and only pay for it while the query is running, depending on your billing options.

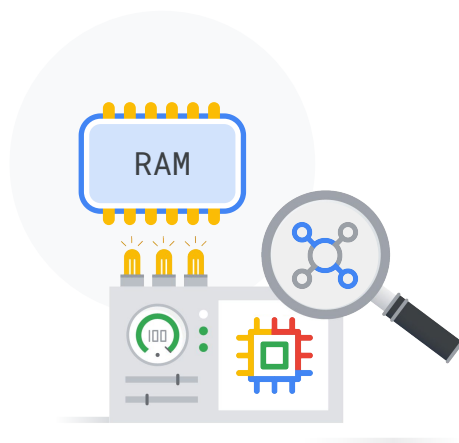
This separation is a game changer for managing costs and ensuring performance.

BigQuery achieves its speed through two core concepts: **slots and shuffle**.

Let's break down how BigQuery achieves its incredible speed by looking at two core concepts: slots and shuffle.

What is a slot?

A small, self-contained unit of computational power that includes **CPU**, **RAM**, and **network bandwidth**.



Think of a slot as a virtual worker—a small, self-contained unit of computational power that includes CPU, RAM, and network bandwidth.

Slots

- 01 BigQuery's Dremel engine assigns thousands of slots to a query job.
- 02 Each slot processes a small piece of data simultaneously.
- 03 This massively parallel processing enables BigQuery to scan terabytes of data quickly.



When you run a query, BigQuery's Dremel engine assigns potentially thousands of these slots to your job. Each slot processes a small piece of your data simultaneously.

This is the "massively parallel processing" that allows BigQuery to scan terabytes of data so quickly.

What is shuffle?

The process of redistributing the intermediate data that the slots have processed.



When the results from all those parallel workers need to be combined, such as for a "GROUP BY" or a "JOIN," the shuffle comes in.

Shuffle is the process of redistributing the intermediate data that the slots have processed.

Shuffle

- 01 Gathers and reorganizes data.
- 02 Sends it to the next set of slots for further processing, such as aggregation or joining.
- 03 Executes complex analytical queries efficiently at a massive scale.



Using Google's petabit internal network, Jupiter, shuffle gathers and reorganizes this data, sending it to the next set of slots for further processing, such as aggregation or joining.

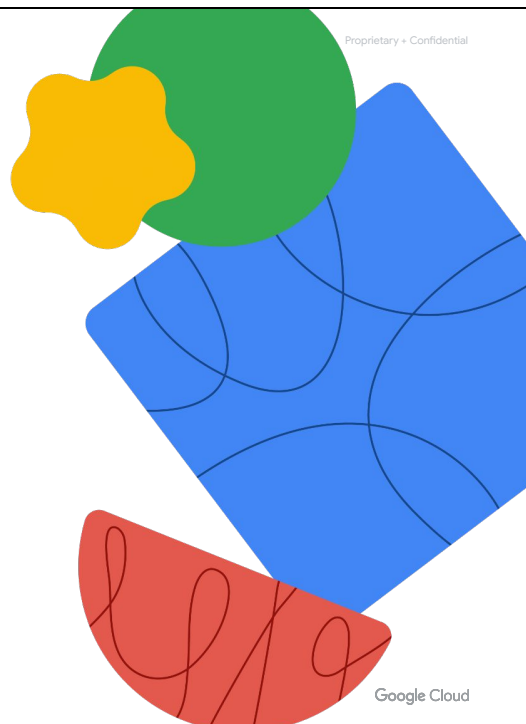
This incredibly fast redistribution of data between query stages is essential for executing complex analytical queries efficiently at a massive scale.

BigQuery's compute and storage
separation enables versatile **multisource**
data querying.

BigQuery's groundbreaking separation of compute and storage is the key to its versatility in querying data from multiple sources.

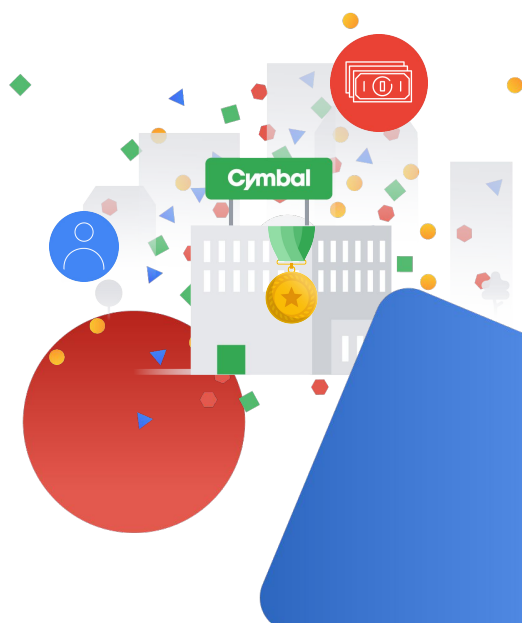
Partitioning and clustering

Two powerful concepts you can use to allow BigQuery to process your data efficiently and cost-effectively.



Now that you understand the basics of BigQuery, you'll learn how to make it even more efficient.

Over the next few slides, we will discuss partitioning and clustering as two powerful concepts you can use to allow BigQuery to process your data efficiently and cost-effectively.



“

Running queries that scan every single piece of data can become slow and expensive, even for BigQuery. You can optimize this with **partitioning and clustering**.



For a company, such as Cymbal, with massive datasets, running queries that scan every single piece of data can become slow and expensive, even for BigQuery.

You can optimize this with partitioning and clustering.

Partitioning

Partitioning is like organizing data into separate drawers for year, month, and day. In BigQuery, you can partition a table based on a date or an integer column.



Partitioning is like organizing data into separate drawers for year, month, and day. In BigQuery, you can partition a table based on a date or an integer column.

Example: Sales data table

transaction_date	customer_id	product_category	sales_amount
2025-08-01	CUST002	Headsets	52
2025-08-01	CUST002	Keyboards	186
2025-08-02	CUST003	Headsets	465
2025-08-02	CUST005	Headsets	57
2025-08-03	CUST003	Shoes	413
2025-08-03	CUST001	Consoles	347

Let's examine a small sample of sales data from Cymbal.

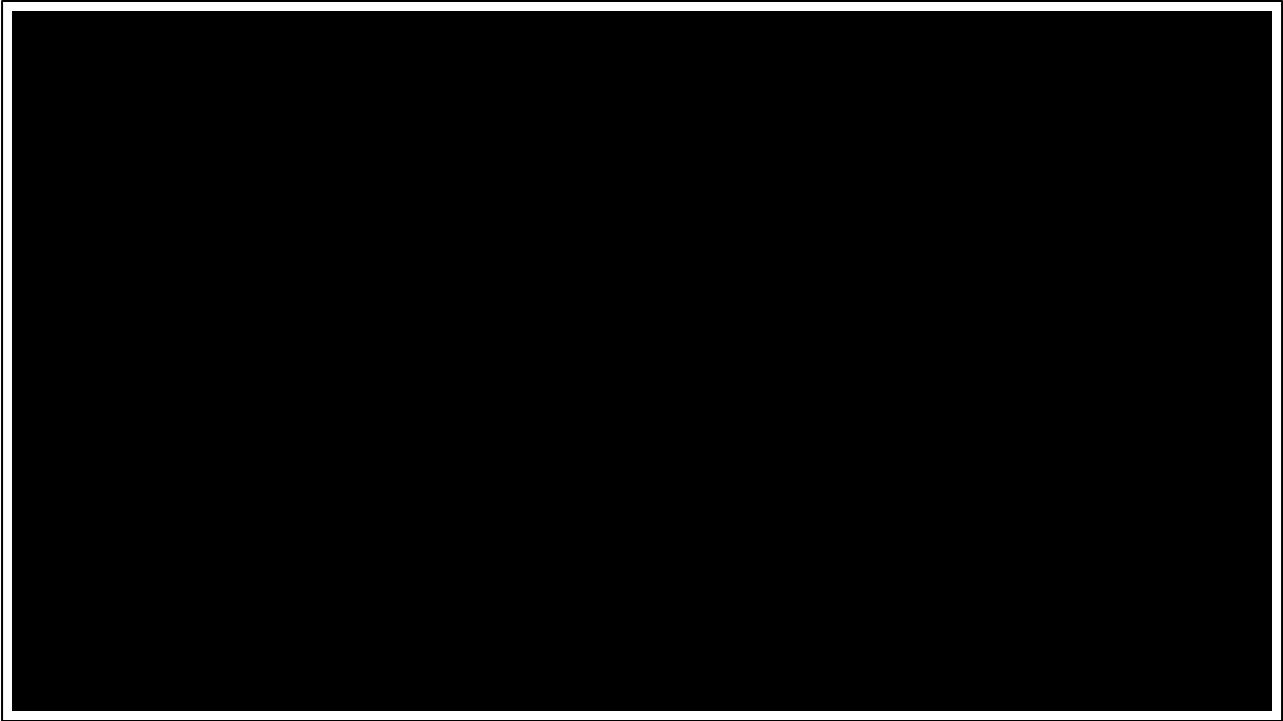
Partitioned by `transaction_date`

transaction_date	customer_id	product_category	sales_amount
2025-08-01	CUST002	Headsets	52
2025-08-01	CUST002	Keyboards	186

transaction_date	customer_id	product_category	sales_amount
2025-08-02	CUST003	Headsets	465
2025-08-02	CUST005	Headsets	57

When you run a query that filters by a specific date range, like for sales from last week, BigQuery knows it only needs to scan the partitions for those specific days.

It completely ignores all the other partitions, which drastically reduces the amount of data scanned. This makes your queries faster and cheaper.



Demo link:

https://drive.google.com/file/d/1yecj4_a00CRA2uYMRr_-Wp891el3OXDz/view?usp=drive_link

Demo guide (demo transcript for the video added at the end):

drive.web-frontend_20251013.10_p0

/*

NOTE: This demo guide is using a different dataset than used in the demo video.
This was done as the video was recorded using a dataset that is not publically
available

and to minimize the resources required to create and run the demo.

*/

-- Using your preferred method, create a dataset named "bq_demo."

-- Run the following query to create the nonpartitioned table.

CREATE TABLE

bq_demo.unpartitioned_taxi

AS

WITH

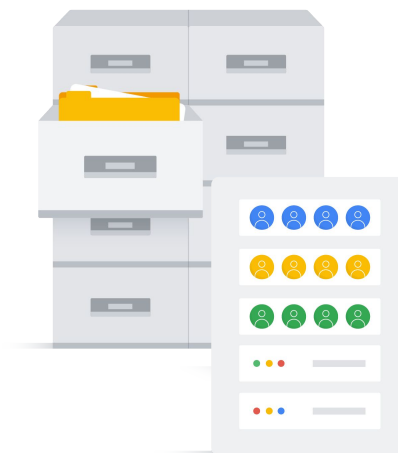
SOURCE AS (

SELECT

DATE(`pickup\_datetime`) AS pickup_date,

Clustering

Clustering sorts the data within each partition. Think of it as organizing the files within each drawer of your filing cabinet alphabetically by customer name.



While partitioning divides the data into large chunks, clustering sorts the data within each of those chunks.

Think of it as organizing the files within each drawer of your filing cabinet alphabetically by customer name.

Clustering example

transaction_date	customer_id	product_category	sales_amount
2025-08-01	CUST002	Headsets	52
2025-08-01	CUST002	Keyboards	186
2025-08-01	CUST003	Headsets	465
2025-08-01	CUST005	Headsets	57
2025-08-01	CUST003	Shoes	413
2025-08-01	CUST001	Consoles	347

For Cymbal, you might cluster a partitioned sales table by "customer_id" or "product_category."

If you run a query to find all purchases made by a specific customer within the last month, BigQuery first goes to the correct monthly partition.

Query example with partition + clustering

```
SELECT *  
FROM cymbal_sales  
WHERE transaction_date = '2025-08-01'  
      AND customer_id = 'CUST002';
```

When dealing with partitioned and clustered data in BigQuery, always put the partition as the first item in your WHERE clause to minimize the partitions accessed.

Then, because the data is sorted by "customer_id," it can jump directly to the data for that customer instead of reading through the entire partition.

Partitioning in Iceberg

01

Leveraging existing iceberg structures

Intelligently leverages the existing structure defined in the Apache Iceberg table's own metadata.

02

Defining partitions

With partitioning, the partitions are typically defined and written by a data processing engine, such as Apache Spark.

03

Data organization

The Apache Iceberg metadata meticulously tracks which specific data files in Cloud Storage belong to which date.

Google Cloud

1. When querying Apache Iceberg tables in Cloud Storage, BigQuery doesn't implement its own partitioning and clustering. Instead, it intelligently leverages the existing structure defined in the Apache Iceberg table's own metadata, demonstrating the power of open standards.
2. With partitioning, the partitions are typically defined and written by a data processing engine, such as Apache Spark.
3. **For instance, a sales data table might be partitioned by transaction_date.** The Apache Iceberg metadata meticulously tracks which specific data files in Cloud Storage belong to which date.

Query filtering example

```
WHERE transaction_date = '2025-08-15'
```

01

BigQuery (via Lakehouse) reads this metadata and identifies the exact set of files corresponding to that date and prunes away all others.

02

Ignores irrelevant data, drastically reducing the amount of data scanned, which lowers costs and accelerates query performance.

When you run a query in BigQuery with a filter, such as `WHERE transaction_date = "2025-08-15,"` BigQuery (via Lakehouse) first reads this metadata.

It instantly identifies the exact set of files corresponding to that date and prunes away all others.

This tells the query workers to ignore irrelevant data, drastically reducing the amount of data scanned, which lowers costs and accelerates query performance.

Clustering in Apache Iceberg

01

Clustering in BigQuery is mirrored by data sorting and file-level statistics in Apache Iceberg.

02

The data within Apache Iceberg's Parquet or ORC files is often sorted by specific columns, such as `customer_id`.

03

The Apache Iceberg metadata stores statistics, such as the minimum and maximum `customer_id` values, for each individual data file.

The concept of clustering in BigQuery is mirrored by data sorting and file-level statistics in Apache Iceberg.

The data within Apache Iceberg's Parquet or ORC files is often sorted by specific columns, such as `customer_id`.

The Apache Iceberg metadata then stores statistics, such as the minimum and maximum `customer_id` values, for each individual data file.

Predicate pushdown

01 If your BigQuery query filters for a specific customer, the query planner consults these statistics.

02 Skips reading any files whose min/max range doesn't contain the requested `customer_id`.

03 Predicate pushdown provides a finer-grained level of data pruning, ensuring maximum query efficiency.

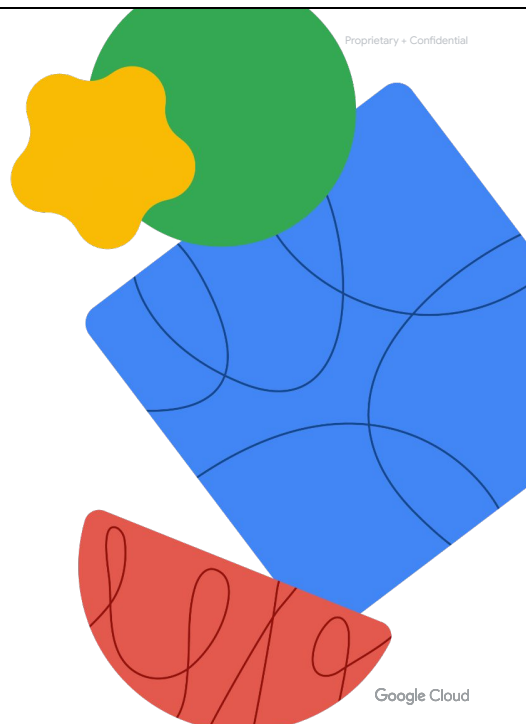
If your BigQuery query filters for a specific customer, the query planner consults these statistics.

It can then skip reading any files whose minimum/maximum range doesn't contain the requested `customer_id`, even if they are in the correct partition.

This powerful predicate pushdown provides a finer-grained level of data pruning, ensuring maximum query efficiency on your open data lakehouse.

Why it matters

Partitioning and clustering together give BigQuery fine-grained control over how your data is queried, leading to significant performance improvements and cost savings, especially as your data grows.



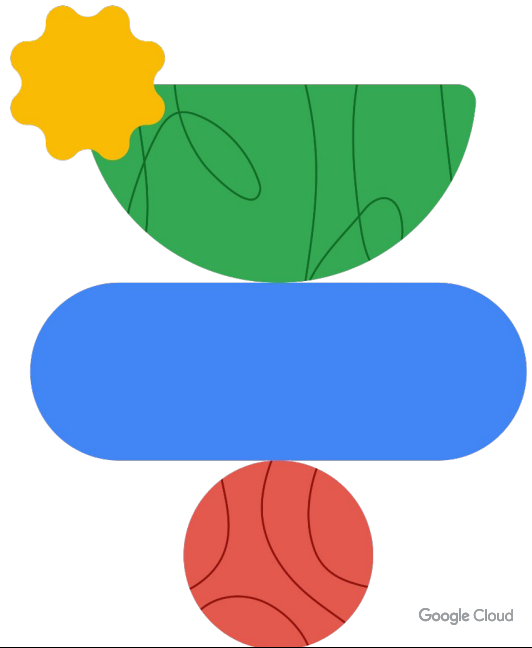
Google Cloud

Partitioning and clustering together give BigQuery fine-grained control over how your data is queried, leading to significant performance improvements and cost savings, especially as your data grows.

BigQuery uses metadata to skip unnecessary data, so **queries run faster** and cost less.

Whether you're working with BigQuery native tables or Apache Iceberg tables, the principle is the same: **BigQuery uses metadata to skip unnecessary data, so queries run faster and cost less.**

**Valuable data exists in
many places and
formats**



You've learned how BigQuery acts as a highly scalable and efficient cornerstone for your modern data warehouse.

However, the reality for most enterprises, including our example company Cymbal, is that valuable data exists in many places and formats.

Data warehouse vs. data lake

Traditional data storage

- Data warehouse: Ideal for structured, organized data used for business insights.
- Data lake: A cost-effective place for huge amounts of raw data in any format.

Challenges of separation

- Data silos: Hard to analyze different data types together.
- Complex extract, transform, load (ETL): Required moving and duplicating data, increasing costs and delays.
- Governance Issues: Managing security and access across two systems was complicated.

Historically, enterprise data was managed in two distinct ways:

- Data warehouse: The traditional standard for decades, perfect for structured, curated data used for business intelligence.
- Data lake: Emerging around 2010 as low-cost repository for storing vast amounts of raw data in any format.

This separation, although logical, often leads to significant challenges. It creates data silos, making it difficult to analyze different types of data together.

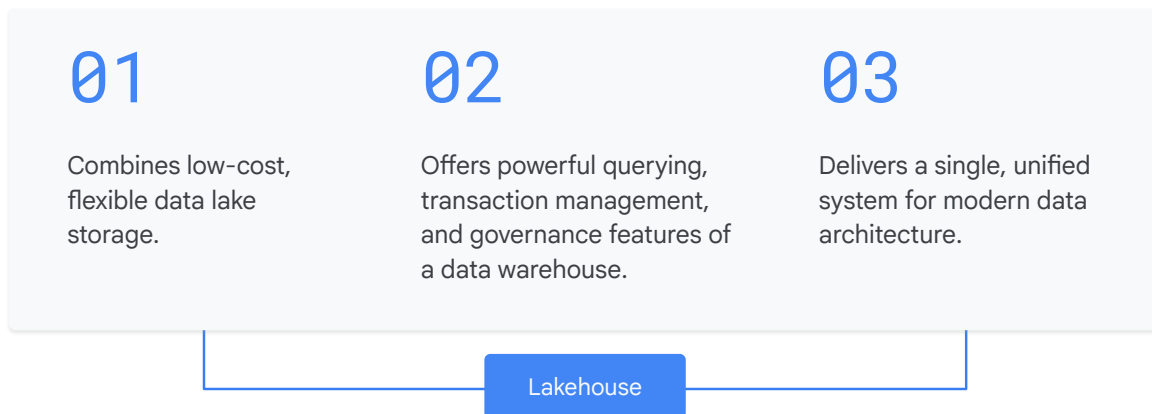
It requires building complex extract, transform, load (ETL) pipelines to move and duplicate data from the data lake to the data warehouse, which introduces latency and increases costs.

It also complicates data governance, as you have to manage security and access policies across two different systems.

A data lakehouse **solves these challenges.**

To solve these problems, a new architectural pattern has emerged: the data lakehouse.

Data lakehouse architecture



A data lakehouse architecture combines the key benefits of both worlds:

The low-cost, flexible storage of a data lake with the powerful querying, transaction management, and governance features of a data warehouse, all within a single, unified system.

On Google Cloud, the service that makes this powerful architecture a reality is Lakehouse.

Lakehouse and external tables

01

Lakehouse extends BigQuery.

02

Create Lakehouse tables over data in object stores, such as Google Cloud Storage.

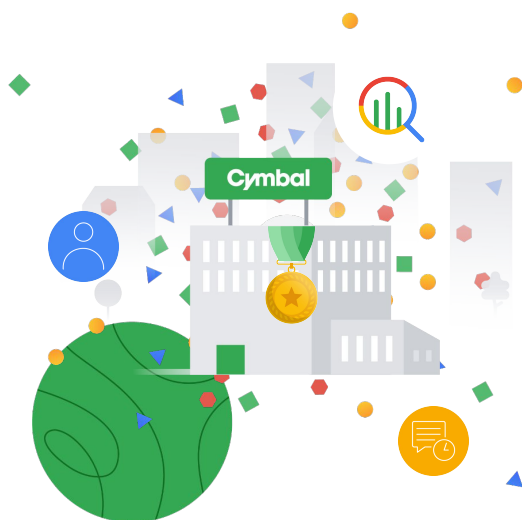
03

Reference the source data files in your data lake, and query them.



Lakehouse extends BigQuery by enabling you to create Lakehouse tables over data in object stores, such as Cloud Storage.

These are external tables that don't store data within BigQuery; instead, they reference the source data files in your data lake, allowing you to query them with unified governance and performance.



How can Cymbal use Lakehouse external tables?

- Cymbal's data science team uses Cloud Storage to store raw web server log files in JSON format.
- Analyzing this clickstream data used to require a complex data pipeline to parse and load it into BigQuery.

Lakehouse simplifies this process.

Google Cloud

- Cymbal's data science team stores raw, semi structured web server log files in JSON format in a Cloud Storage bucket.
- To analyze this clickstream data, they would historically need a complex data pipeline to parse and load the data into BigQuery.
- With Lakehouse, the process is much simpler.

Lakehouse external table

01

They create a Lakehouse table directly over the JSON files in Cloud Storage.

02

Use the familiar BigQuery SQL interface to query clickstream data instantly.

03

Join this external table with a native BigQuery sales table, to discover correlations between website behavior and purchasing habits.

Cymbal can create a Lakehouse table directly on top of the JSON files in Cloud Storage. Now, you can use the familiar BigQuery SQL interface to query this clickstream data instantly, as if it were a native BigQuery table.

You can even join this external table with a native BigQuery sales table to discover correlations between website behavior and purchasing habits, without ever moving or duplicating data.

Lab: Introduction to querying Apache Iceberg tables with BigQuery

Overview

Explore how BigQuery's Lakehouse tables can query data in open-source formats directly within Cloud Storage.



What you'll do

- Create a Lakehouse table that points to an existing Apache Iceberg table.
- Analyze data without moving or duplicating it.

This lab will introduce you to querying Apache Iceberg tables with BigQuery. You will explore how BigQuery Lakehouse tables can query data in open-source formats directly within Cloud Storage.

What you'll do:

- Create a Lakehouse table that points to an existing Apache Iceberg table.
- Analyze data without moving or duplicating it.

MODULE 10

Advanced Lakehouse Patterns and Data Governance

· 14:00–14:35 ·

instructor slides 6–9, 11–13, 20–25, 43–48 (medallion), 49–53
(student pages remapped +3 after p40 — student deck has 3 extra demo-transcript pages)

source deck: M4_Advanced_Lakehouse_Patterns_and_Data_Governance.pdf

What is metadata?

Metadata is data about data, which can include:

01

Who created the data

04

How it relates to other data

02

When it was created

05

Who owns it

03

What it contains

06

Its security sensitivity

Metadata is data about data.

It can include

- Who created the data
- When it was created
- What it contains
- How it relates to other data
- Who owns it
- Its security sensitivity

Knowledge Catalog for Cymbal

Without a centralized metadata system,
data management can be difficult.
Knowledge Catalog provides a unified
metadata hub.



Google Cloud

Without a centralized metadata system, data management can be difficult. Dataplex
Universal Catalog provides a unified metadata hub.

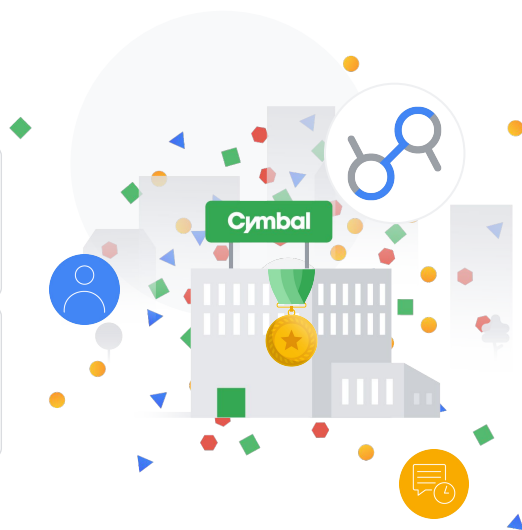
Knowledge Catalog for Cymbal

01

Acts as a universal catalog for all data assets, whether they reside in BigQuery, Cloud Storage, or Lakehouse.

02

Use to discover data, understand data lineage, and manage and augment metadata.



Google Cloud

For Cymbal, Knowledge Catalog acts as a universal catalog for all data assets, whether they reside in BigQuery, Cloud Storage, Google Cloud Lakehouse (Lakehouse), or other supported systems.

For Cymbal's data analysts, this centralized catalog is critical.

Instead of searching through different systems to find the required datasets, they use the Knowledge Catalog to discover data, understand data lineage, and manage and augment metadata.

By providing a single reference source,
Knowledge Catalog helps Cymbal manage
data at scale while **ensuring consistency
and quality**.

By providing a single reference source, Knowledge Catalog helps Cymbal manage data at scale while ensuring consistency and quality.

Sensitive Data Protection



Protect customers' sensitive information, such as names, addresses, and credit card numbers.



A data breach can damage Cymbal's reputation and lead to significant financial penalties.



Sensitive Data Protection is an essential tool for securing data.

Cymbal has a responsibility to protect its customers' sensitive information, such as names, addresses, and credit card numbers.

A data breach can damage their reputation and lead to significant financial penalties.

Sensitive Data Protection is an essential tool for this purpose.

Sensitive Data Protection

Discover

Run scans on data lakehouse and other data to identify multiple types of sensitive field.

Classification

Data is classified by sensitivity level. This ensures the right security controls are applied.

Protection

Use techniques, such as masking or tokenization, to de-identify the data.

Example: *****9787

Sensitive Data Protection allows Cymbal to automatically discover, classify, and protect sensitive data across its data lakehouse. Let's learn more about each function.

- **Discovery:** Cymbal runs scans on BigQuery tables and Cloud Storage buckets to identify sensitive data. For example, they can configure a scan to look for patterns that match credit card numbers or email addresses.
- **Classification:** Once identified, data is classified by its sensitivity level. This ensures the right security controls are applied.
- **Protection:** To protect the data, Cymbal can use techniques, such as masking or tokenization, to de-identify it. For instance, a customer support representative might see only the last four digits of a credit card number, while the full number is replaced with a nonsensitive token.

This proactive approach to data protection
allows Cymbal to **use their data for
analytics and ML** without exposing
sensitive information.

This proactive approach to data protection allows Cymbal to use their data for analytics and ML without exposing sensitive information.

Fine-grained security

Column-level security

Restricts access to specific columns in a table. For example, a marketing analyst might be able to access a customer's purchase history but not their contact information. This is effective for protecting Personally Identifiable Information (PII).

Row-level security

Filters which rows a user can access. A regional sales manager for North America, for instance, would only have access to sales data for that region. This is particularly useful for large, multinational companies, such as Cymbal.

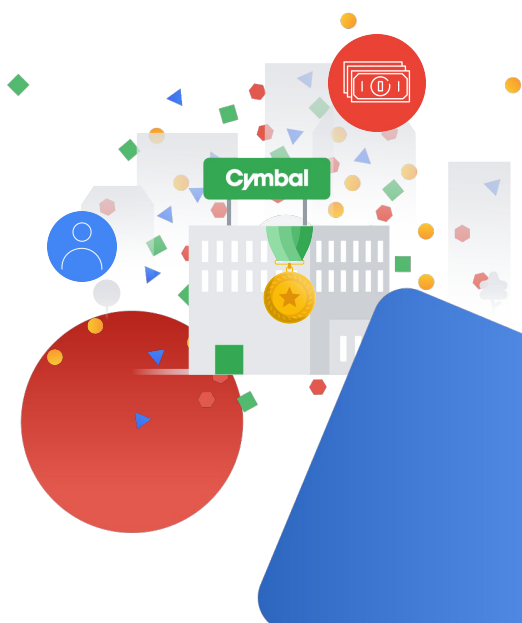
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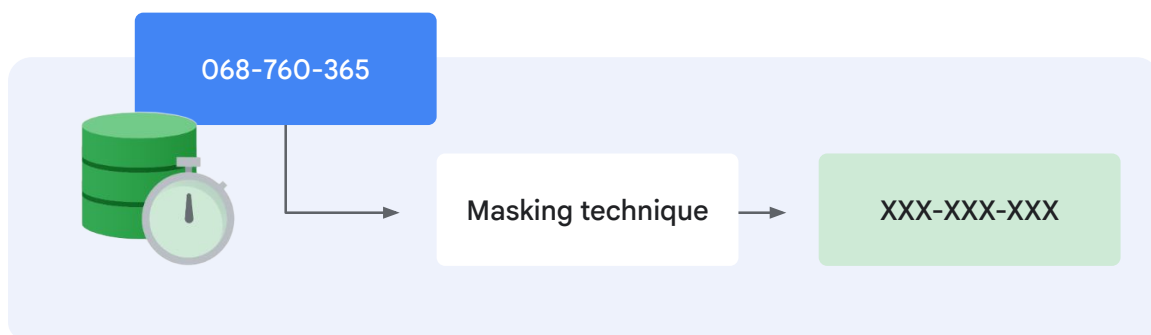


“

Cymbal can ensure that their data is secure and accessible to the correct people at the appropriate time.

By combining IAM with these fine-grained controls, Cymbal can ensure that their data is secure and accessible to the correct people at the appropriate time.

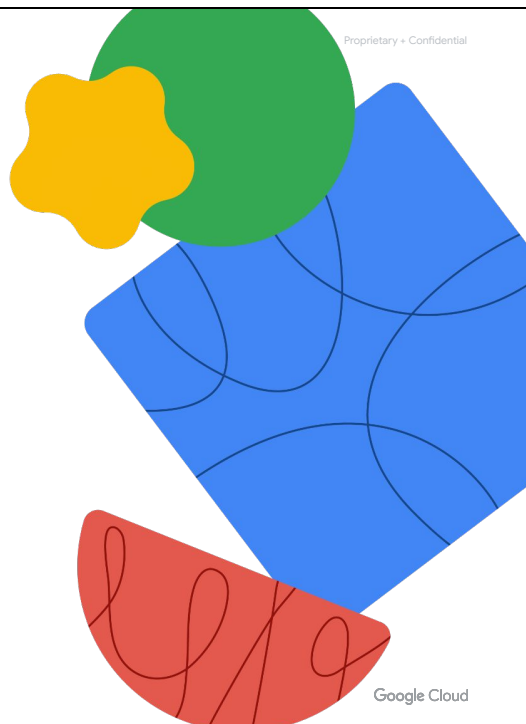
For Lakehouse tables in Cloud Storage, Sensitive Data Protection can apply **dynamic data masking**.



With Sensitive Data Protection, data can be de-identified and protected using dynamic data masking, redaction, tokenization, date shifting, and more.

Data loss prevention

Demonstration



Google Cloud

Next, we'll go over a demonstration of how to use Sensitive Data Protection to find PII in a dataset.

Scenario: Cymbal loyalty program

01

New customer information has been collected and loaded into a BigQuery table named `loyalty_program_customers`.

02

This table contains standard information, such as names and purchase histories.

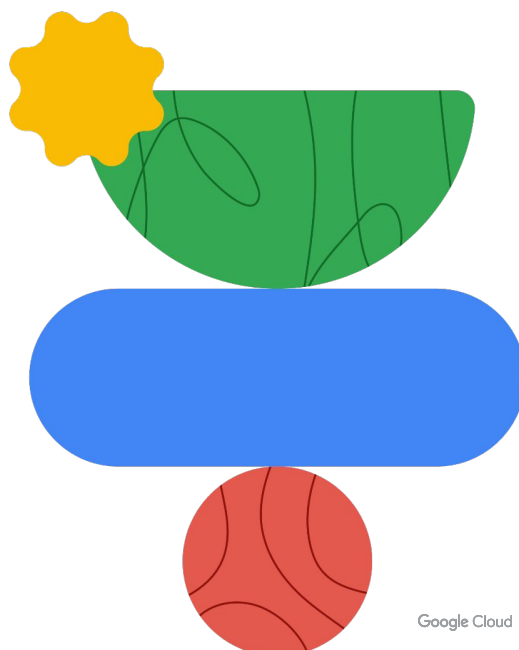
03

Includes potentially sensitive data, such as email addresses, phone numbers, and free-text comments from customer feedback surveys.

In this scenario, Cymbal has launched a new loyalty program. New customer information has been collected and loaded into a BigQuery table named `loyalty_program_customers`.

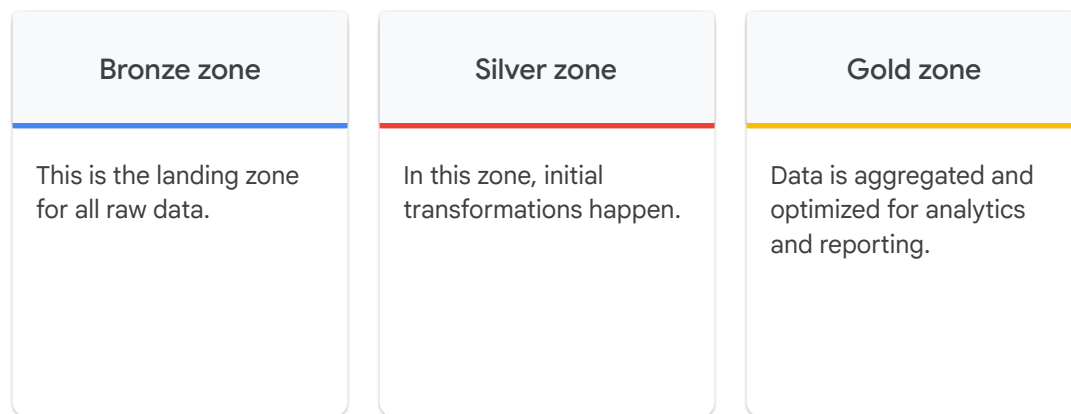
This table contains standard information, such as names and purchase histories, and also includes potentially sensitive data, such as email addresses, phone numbers, and free-text comments from customer feedback surveys.

Before this data is made available to the marketing analytics team, **any PII must be properly handled** to avoid exposing sensitive customer details in analytics dashboards.



Before this data is made available to the marketing analytics team, any PII must be properly handled to avoid exposing sensitive customer details in analytics dashboards.

The medallion architecture (1 of 3)



The medallion architecture organizes data into three distinct zones: bronze, silver, and gold.

- **Bronze zone (raw data):** This is the landing zone for all raw data.
- **Silver zone (cleansed and conformed data):** In this zone, the data is cleaned, validated, and enriched. This is where initial transformations happen.
- **Gold zone (curated business-level data):** This is the final, highly refined layer. The data here is aggregated and optimized for analytics and reporting. This data almost always resides in native BigQuery tables for maximum query performance.

Cymbal: Bronze zone (raw data)

01

Clickstream data from their website.

02

Batch exports of transactional data from their e-commerce database.

03

JSON data from their social media marketing campaigns.

For Cymbal, the bronze zone includes

- Clickstream data from their website, streamed in real-time through Pub/Sub and landing in a Cloud Storage bucket.
- Batch exports of transactional data from their e-commerce database, saved as CSV or Avro files in Cloud Storage.
- JSON data from their social media marketing campaigns.

Cymbal: Silver zone (cleansed and conformed data)

01

Parsed clickstream data with structured sessions.

02

Augmented transactional data.

03

Standardized date and time formats.

The silver zone includes

- Parsing the raw clickstream data to create structured sessions.
- Joining the transactional data with customer dimension tables.
- Standardizing date and time formats.

Cymbal: Gold zone (curated business-level data)

01

Aggregated daily sales tables.

02

Customer 360-degree view tables.

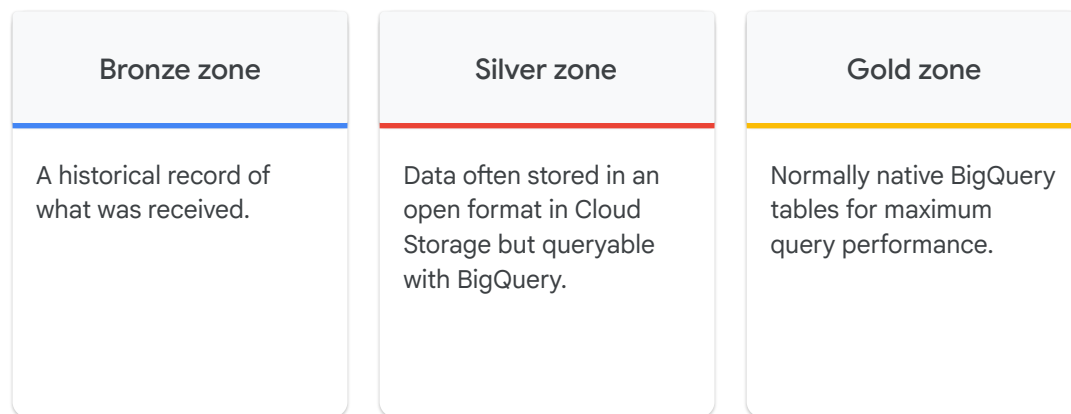
03

Inventory performance summaries.

Finally, the gold zone includes

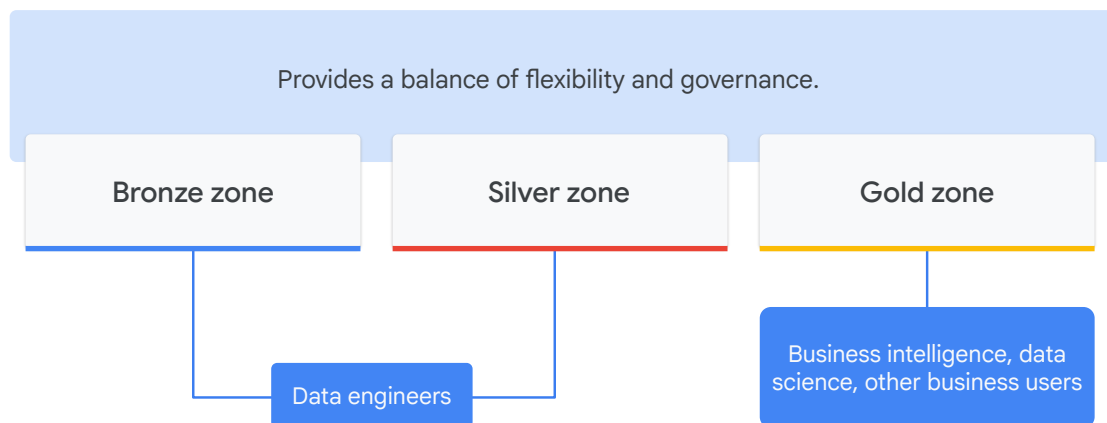
- Aggregated daily sales tables.
- Customer 360-degree view tables.
- Inventory performance summaries.

The medallion architecture (2 of 3)



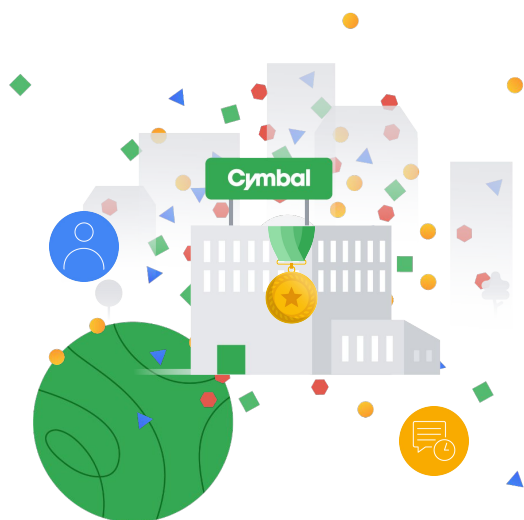
- Data in the bronze zone is normally immutable—it's a historical record of what was received.
- Data in the silver zone is often stored in an open format, such as Parquet, or as Lakehouse tables, making it queryable through BigQuery but still residing in Cloud Storage.
- Data in the gold zone almost always resides in native BigQuery tables for maximum query performance.

The medallion architecture (3 of 3)



This layered approach provides a balance of flexibility and governance.

Data engineers can work with the raw data in the bronze and silver zones, whereas business intelligence tools, data science teams, and other business users get fast, reliable access to the curated data in the gold zone.



Cymbal migration strategies

For a company, such as Cymbal, which might be running on a traditional on-premises data warehouse (such as Teradata or Hadoop), migrating to a cloud-native data lakehouse is a significant project.

A complete, one-time migration is often too risky and disruptive. Instead, a phased, use-case-driven approach is usually more successful.

For a company, such as Cymbal, which might be running on a traditional on-premises data warehouse (such as Teradata or Hadoop), migrating to a cloud-native data lakehouse is a significant project.

A complete, one-time migration is often too risky and disruptive. Instead, a phased, use-case-driven approach is usually more successful.

Migration strategy

- 01 Set up the core infrastructure on Google Cloud
- 02 Start with a high-impact use case
- 03 Migrate relevant data
- 04 Build the new pipelines and reports
- 05 Decommission and iterate

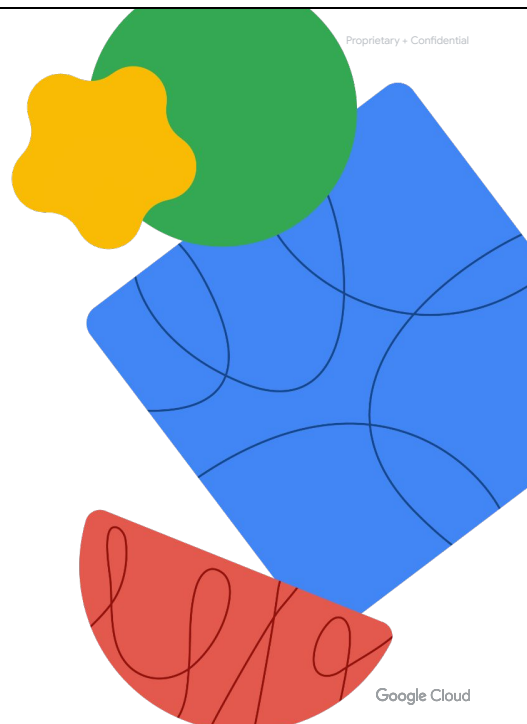


Google Cloud

1. **Set up the core infrastructure on Google Cloud:** This includes setting up a Google Cloud project with the appropriate IAM permissions, creating Cloud Storage buckets for the bronze, silver, and gold zones, and setting up Knowledge Catalog to manage metadata.
2. **Start with a high-impact use case:** Instead of migrating everything at once, Cymbal could pick one specific business problem to solve, such as marketing analytics. This is an area where having access to both structured and unstructured data can provide significant value.
3. **Migrate relevant data:** This could involve using the BigQuery Data Transfer Service to set up recurring transfers from their on-premises warehouse and using a tool, such as Dataflow, to ingest new, real-time data into their Cloud Storage bronze zone.
4. **Build the new pipelines and reports:** With the data flowing into Google Cloud, they can start building the new data pipelines to populate their silver and gold zones. The marketing team can then build new dashboards in a tool, such as Looker, pointing to the new gold tables in BigQuery.
5. **Decommission and iterate:** Once the new solution is running successfully and approved by business users, they can decommission the old on-premises reports. This demonstrates value and builds momentum for the next phase of the migration. They can then repeat this process for other use cases.

Cost management and optimization best practices

A proactive approach to cost management.



Google Cloud

A key benefit of the cloud is the pay-as-you-go model, but it also requires a proactive approach to cost management.

Let's review some best practices that Cymbal would implement.

Best practices (1 of 2)

01

Choose the right storage class

Not all data needs to be accessed with the same frequency. Consider storage class in Cloud Storage.

02

Optimize BigQuery queries

Use features, such as partitioning and clustering, as well as writing efficient SQL queries.

1. Choose the right storage class: Not all data needs to be accessed with the same frequency.
For the raw data in their bronze zone, which might be accessed infrequently, they can use a cheaper storage class, such as Nearline or Coldline, in Cloud Storage.
2. Optimize BigQuery queries: They can train their analysts to write efficient SQL queries.
BigQuery provides tools to estimate the cost of a query before it's run. They can also use features, such as partitioning and clustering their tables, to reduce the amount of data scanned by each query.

Best practices (2 of 2)

03

Use BigQuery's flat-rate pricing

Great for predictable workloads.

04

Set up budgets and alerts

In Google Cloud's billing tools.

Google Cloud

1. Use BigQuery's flat-rate pricing: For predictable workloads, Cymbal can switch from on-demand pricing to a flat-rate model, where they purchase dedicated query processing capacity at a fixed monthly cost.
2. Set up budgets and alerts: In Google Cloud's billing console, they can set up budgets for their projects and create alerts that notify them when costs are approaching their limits.

MODULE 11

BQML + the Power BI Last Mile (compressed)

· 14:50–15:20 ·

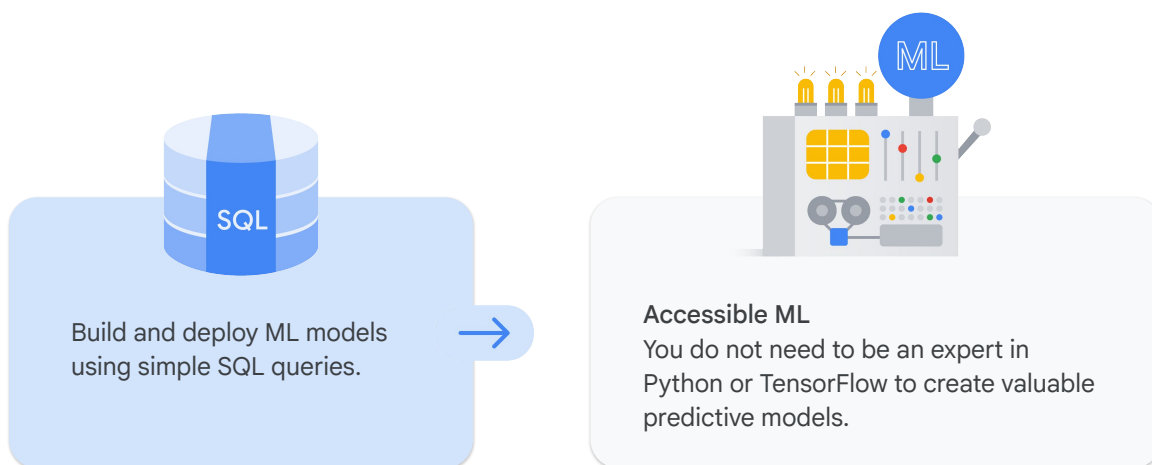
M4 (inst. slides 30–37): BigQuery ML

source deck: M4_Advanced_Lakehouse_Patterns_and_Data_Governance.pdf

M5 slides 5–20: recap, best practices, AI direction

source deck: M5_Labs_and_best_practices.pdf

BigQuery ML



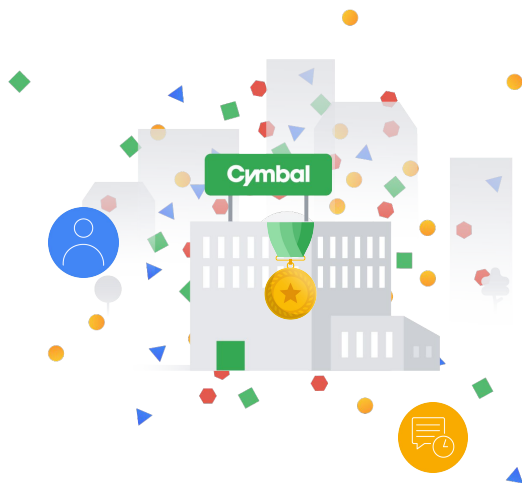
One of the most powerful tools in the Google Cloud analytics toolkit is BigQuery ML.

It allows data analysts and data scientists at Cymbal to build and deploy ML models using simple SQL queries.

This makes ML accessible to more people, so you don't need to be an expert in Python or TensorFlow to create valuable predictive models.

Use case

To identify customers who are at risk of not making another purchase, the marketing analytics team can use BigQuery ML and a few SQL statements.



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Use case: Predicting customer churn (1 of 4)

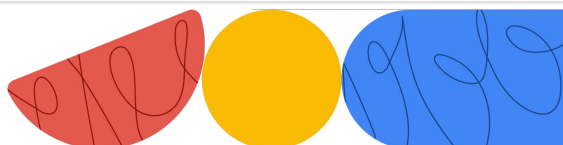
01

Feature engineering

The first step is to prepare the data. The analysts write a SQL query to create features (or signals) that might predict churn.

Fields

- `recency`: days since last purchase
- `frequency`: purchases in the last year
- `monetary_value`: total spent
- `days_since_first_purchase`: customer tenure



The first step is to prepare the data. The analysts write a SQL query to create features (or signals) that might predict churn.

These could include the following fields:

- *recency*: days since last purchase
- *frequency*: purchases in the last year
- *monetary_value*: total spent
- *days_since_first_purchase*: customer tenure

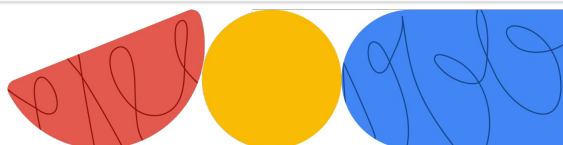
Use case: Predicting customer churn (2 of 4)

02

Model training

After the features are ready, they train a model with a single `CREATE MODEL` statement in SQL.

For this binary classification problem (churn or no churn), a logistic regression or a boosted tree model is a good choice.



Model training: After the features are ready, they train a model with a single `CREATE MODEL` statement in SQL.

For this binary classification problem (churn or no churn), a logistic regression or a boosted tree model is a good choice.

Sample code

```
SQL
CREATE OR REPLACE MODEL cymbal_ecommerce.customer_churn_predictor
OPTIONS(model_type='LOGISTIC_REG') AS
SELECT
  customer_id,
  recency,
  frequency,
  monetary_value,
  (total_purchases > 1) AS will_return -- This is our label
FROM
  cymbal_ecommerce.customer_purchase_summary;
```

BigQuery handles all the complexities of model training behind the scenes.

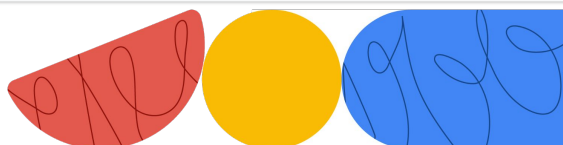
Use case: Predicting customer churn (3 of 4)

03

Model evaluation

After the model is trained, the analysts evaluate its performance using the `ML.EVALUATE` function.

- ✓ Accuracy
- ✓ Precision
- ✓ Recall



After the model is trained, the analysts evaluate its performance using the `ML.EVALUATE` function.

This provides metrics, such as accuracy, precision, and recall, helping them understand how well the model is performing.

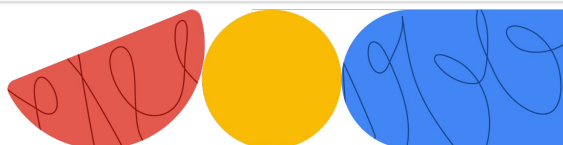
Use case: Predicting customer churn (4 of 4)

04

Prediction

With the `ML.PREDICT` function, you can get a list of all customers and their probability of churning.

This list can then be used to create targeted marketing campaigns to re-engage at-risk customers.



The final step is to use the model to make predictions on new data.

With the `ML.PREDICT` function, they can get a list of all customers and their probability of churning.

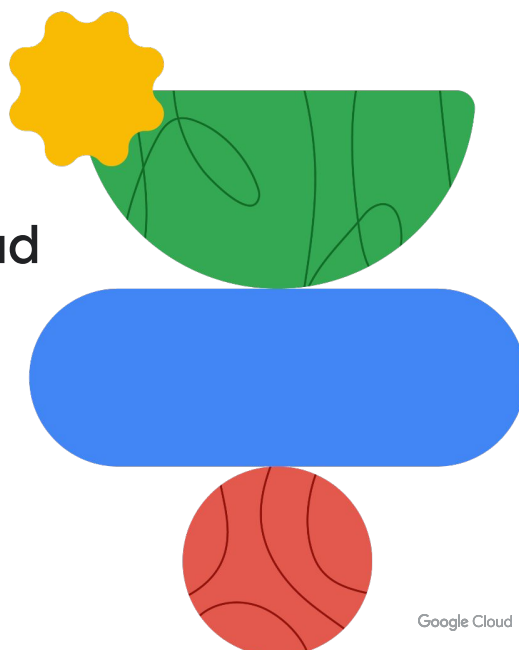
This list can then be used to create targeted marketing campaigns to re-engage at-risk customers.

BigQuery ML allows the teams who are closest to the data to **build predictive models**, which leads to faster and more relevant insights.

For Cymbal, BigQuery ML is a powerful tool. It allows the teams who are closest to the data to build predictive models, which leads to faster and more relevant insights.

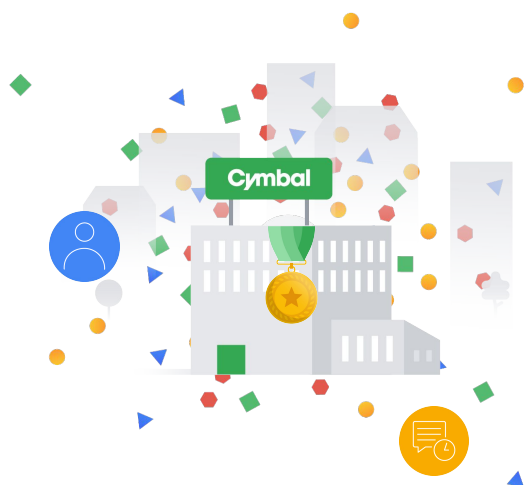
Building a modern data lakehouse on Google Cloud

Leveraging powerful,
open-source-based services.



Google Cloud

This lesson will review building a modern data lakehouse on Google Cloud by leveraging its powerful, open-source-based services.



Cymbal deals with a massive amount of data daily in various formats. A data lakehouse architecture helps bring it together for powerful analytics and AI-powered customer experiences.

Cymbal deals with a massive amount of data every day.

From customer clicks on their website and mobile app to supplier inventory feeds and sales transactions, the data comes in various formats.

A data lakehouse architecture allows Cymbal to bring all this data together for powerful analytics and, increasingly, for building innovative AI-powered customer experiences.

Core components of a Google Cloud data lakehouse

A robust and scalable data lakehouse on Google Cloud is built on a foundation of several key services that work together seamlessly.



Google Cloud

A robust and scalable data lakehouse on Google Cloud is built on a foundation of several key services that work together seamlessly. Let's review what these are.

Cloud Storage

This is the heart of your data lakehouse, providing a highly scalable and cost-effective object storage service. Think of it as the foundational layer where all your data, in its raw and processed forms, resides.

Cymbal's data

- ✓ Website clickstream data
- ✓ Product images
- ✓ Supplier inventory files (CSV/Parquet)
- ✓ Historical data (Apache Iceberg)

This is the heart of your data lakehouse, providing a highly scalable and cost-effective object storage service.

Think of it as the foundational layer where all your data, in its raw and processed forms, resides.

For Cymbal, this means storing everything from website clickstream data in JSON format to product images, supplier inventory files in CSV or Parquet, and historical data in Apache Iceberg format.

BigQuery

This powerful, serverless data warehouse acts as the analytics engine for your data lakehouse. What makes BigQuery unique is its ability to directly query data in Cloud Storage without needing to move it.

The separation of storage and compute is a core principle. BigQuery provides:

- ✓ Flexibility
- ✓ Cost savings

This powerful, serverless data warehouse acts as the analytics engine for your data lakehouse.

What makes BigQuery unique is its ability to directly query data in Cloud Storage without needing to move it.

This separation of storage and compute is a core principle, offering flexibility and cost savings.

Knowledge Catalog

As your data grows, managing it can become a significant challenge. Knowledge Catalog provides an intelligent data fabric that allows you to discover, manage, monitor, govern, and describe your data across your entire data lakehouse.

Cymbal's data

- Knowledge Catalog automatically catalogs data in Cloud Storage buckets, making it easily searchable for data analysts and scientists.
- It enforces data quality rules and security policies, protecting sensitive customer information.

As your data grows, managing it can become a significant challenge.

Knowledge Catalog provides an intelligent data fabric that allows you to discover, manage, monitor, govern, and describe your data across your entire data lakehouse.

For Cymbal, Knowledge Catalog can automatically catalog the data in their Cloud Storage buckets, making it easily searchable for their data analysts and scientists.

It also helps enforce data quality rules and security policies, ensuring that sensitive customer information is protected.

AlloyDB for PostgreSQL

Although the data lakehouse is excellent for analytical workloads, many applications still rely on transactional databases.

AlloyDB is a fully managed, PostgreSQL-compatible database service that offers superior performance and availability.

Cymbal's data

- AlloyDB for e-commerce backend handles real-time order processing and customer account management.
- Integrate AlloyDB data into data lakehouse for broader analytics via federated queries.

Although the data lakehouse is excellent for analytical workloads, many applications still rely on transactional databases.

AlloyDB is a fully managed, PostgreSQL-compatible database service that offers superior performance and availability.

In Cymbal's architecture, AlloyDB could power their e-commerce platform's backend, handling real-time order processing and customer account management.

The data from AlloyDB can then be easily integrated into the data lakehouse for broader analytics via federated queries.

Apache Iceberg

An open-source table format that brings the reliability and performance of a traditional database to your data lake.

Using Apache Iceberg with BigQuery and Cloud Storage, you can perform efficient updates and deletes on your data, a capability that was traditionally difficult in data lakes.

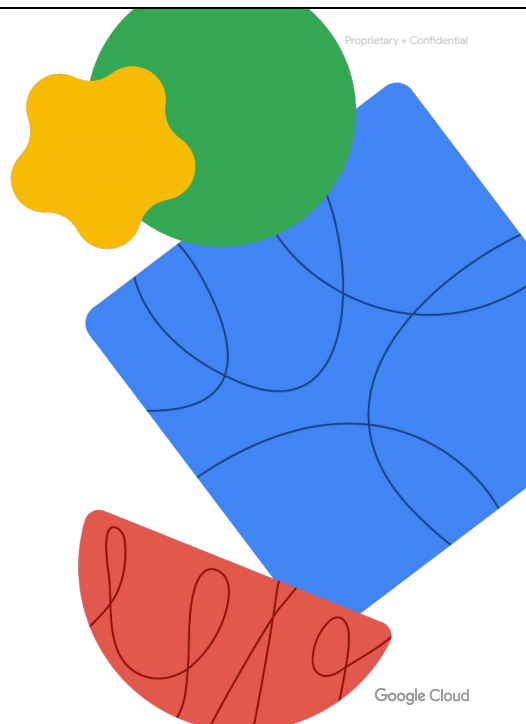
This is crucial for Cymbal when they need to update product information or manage customer data privacy requests.

Apache Iceberg is an open-source table format that brings the reliability and performance of a traditional database to your data lake. Using Apache Iceberg with BigQuery and Cloud Storage, you can perform efficient updates and deletes on your data, a capability that was traditionally difficult in data lakes.

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Architectural patterns and best practices

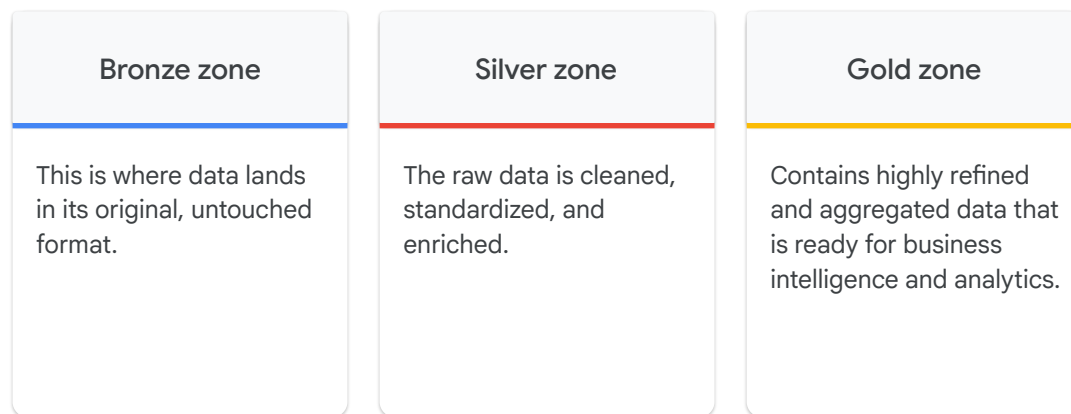
The most common and effective architectural pattern for a data lakehouse is the medallion architecture, which organizes data into three distinct zones.



Google Cloud

You've learned that the most common and effective architectural pattern for a data lakehouse is the medallion architecture, which organizes data into three distinct zones. Let's review.

The medallion architecture



The three zones are

- **Bronze zone (raw)**: This is where data lands in its original, untouched format.
- **Silver zone (cleansed and conformed)**: In this zone, the raw data is cleaned, standardized, and enriched.
- **Gold zone (curated)**: This zone contains highly refined and aggregated data that is ready for business intelligence and analytics.

Best practices

- 01 Embrace open standards
- 02 Govern your data from the start
- 03 Optimize for cost and performance
- 04 Automate your data pipelines

Embrace open standards:

Using open formats, such as Apache Iceberg and Parquet, ensures that you're not locked into a single vendor and can use a variety of tools to work with your data.

Govern your data from the start:

Implementing a data governance framework with tools, such as Knowledge Catalog, is crucial for maintaining data quality, security, and compliance.

Optimize for cost and performance:

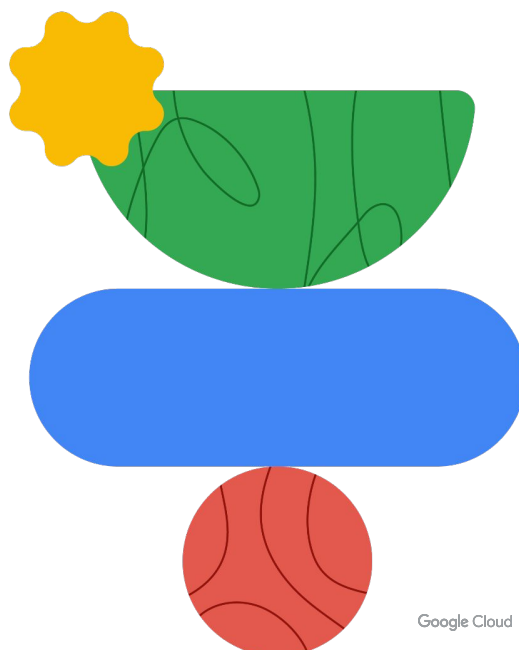
Leverage BigQuery's separation of storage and compute to your advantage. Use partitioning and clustering in your tables to improve query performance and reduce costs.

Automate your data pipelines:

Use tools, such as Dataflow and Managed Service for Apache Spark, to build automated and scalable pipelines for ingesting and transforming your data.

Common pitfalls to avoid

Implementing a data lakehouse is a journey, and there are some common challenges to be aware of.



Implementing a data lakehouse is a journey, and there are some common challenges to be aware of.

Common pitfalls to avoid

- 01 Neglecting governance
- 02 Ignoring data quality
- 03 Vendor lock-in
- 04 Lack of clear strategy

Neglecting governance:

Without proper governance, a data lake can quickly turn into a "data swamp," where it's difficult to find, trust, and use the data.

Ignoring data quality:

Poor data quality can lead to inaccurate analysis and flawed business decisions. Implement data quality checks at every stage of your data pipeline.

Vendor lock-in:

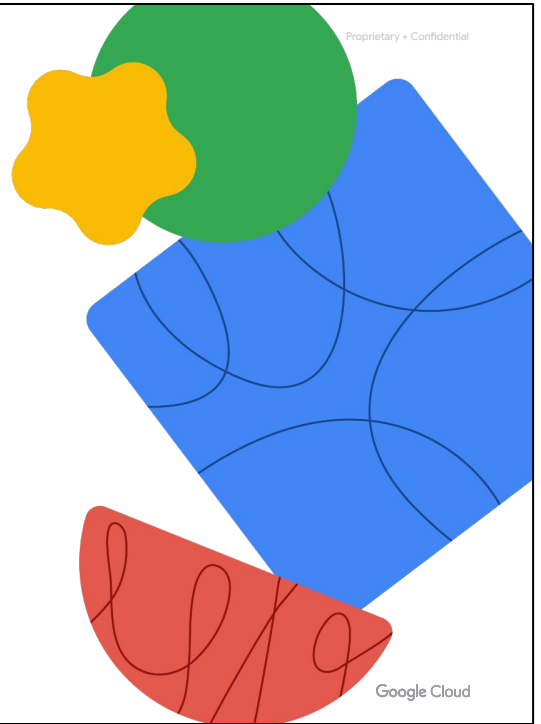
Relying too heavily on proprietary formats and services can limit your flexibility in the future. Prioritize open standards and interoperability.

Lack of clear strategy:

Before you start building, have a clear understanding of your business goals and how the data lakehouse will help you achieve them.

The future: AI and data modernization

The true power of a modern data platform lies in its ability to support advanced analytics and AI.



The true power of a modern data platform lies in its ability to support advanced analytics and AI.

With a well-architected data lakehouse on Google Cloud, you're perfectly positioned to take advantage of the latest trends in data modernization.

AI and data modernization (1 of 2)

01

Advanced analytics can be done quickly using BigQuery ML.

02

Services, such as Agent Platform, can be used to build and train ML models directly on the data in your data lakehouse.

03

Cymbal could develop a recommendation engine for product suggestions or a fraud detection model for suspicious transactions.

Advanced analytics can be done quickly using BigQuery ML.

Services, such as Agent Platform, can be used to build and train ML models directly on the data in your data lakehouse.

For Cymbal, this could mean developing a recommendation engine that suggests products to customers based on their browsing history or a fraud detection model that identifies suspicious transactions in real time.

AI and data modernization (2 of 2)

The rise of generative AI also opens up exciting new possibilities. Cymbal could use large language models to create engaging product descriptions automatically or to power a chatbot that provides personalized customer support.



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COURSE SUMMARY

Services Come Together + Course 1 Key Points

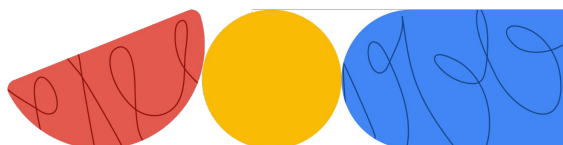
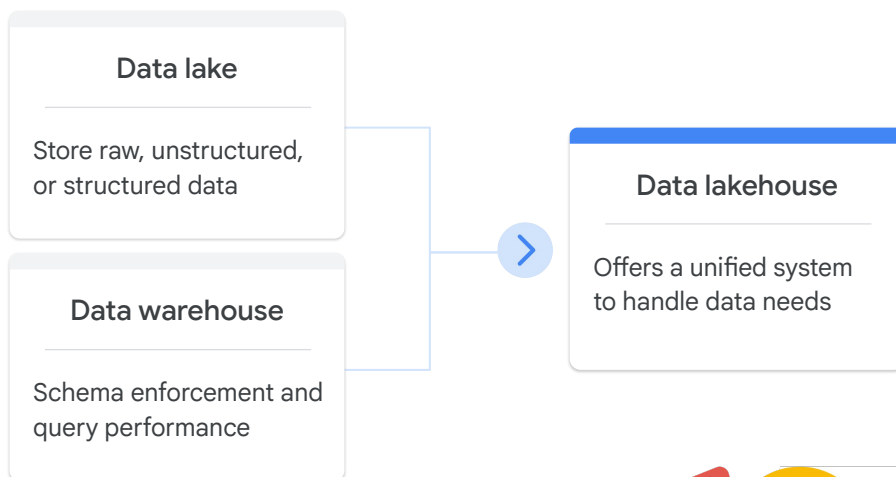
· 15:50–16:10 ·

C2-M6 slides 2–15

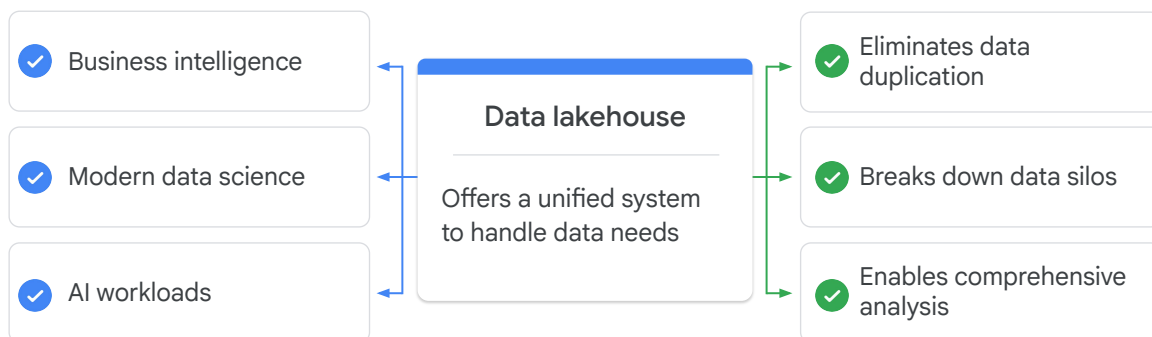
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C1-M7 — all 5 slides

source deck: M7___T-DEGC-B___Course_Summary_ILT_Notes.pdf



A data lakehouse on Google Cloud unifies the strengths of data lakes and data warehouses, combining the flexibility and low cost of a data lake with the speed and management features of a data warehouse.



This creates a single, unified platform for traditional business intelligence, modern data science, and AI workloads, eliminating data duplication.

This approach effectively breaks down data silos, enabling comprehensive analysis of diverse data types that were previously difficult to combine.

How Google Cloud services come together
to **form a modern, efficient, and scalable
data architecture.**

By now, you should understand how Google Cloud services come together to form a modern, efficient, and scalable data architecture.

Now let's go over and reinforce the foundational concepts and practical tools you've learned throughout the modules.

Cloud Storage

The foundational, cost-effective object storage for all raw data, including

- ✓ Structured
- ✓ Semistructured
- ✓ Unstructured
- ✓ Multimodal



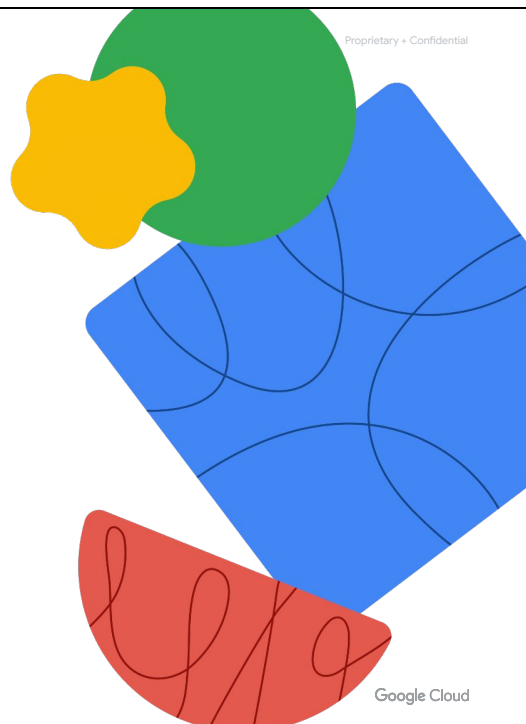
Google Cloud

Cloud Storage is the foundational, cost-effective object storage for all raw data, including structured, semistructured, unstructured, and multimodal formats.

It supports "schema-on-read," allowing immediate data ingestion without predefined structure.

Apache Iceberg

A data lakehouse needs structure and performance. Open table formats, such as Apache Iceberg, provide this.



You also learned about Apache Iceberg, an open table format that adds structure and performance to data residing in Cloud Storage. It organizes data files into queryable tables.

What does Apache Iceberg provide?

01

Schema evolution

Allows schema changes (such as adding/renaming columns) without impacting existing data or applications.

02

Hidden partitioning

Automatically manages how data is partitioned for efficient querying.

03

Time travel

Enables querying past data versions for auditing and debugging, allowing users to reproduce exact data states from previous points in time.

04

Atomic transactions

Enable reliable, concurrent data operations, preventing corruption when multiple processes read/write to the same tables.

It organizes data files into queryable tables, offering critical features, such as

- Schema evolution (adapting to changing data requirements).
- Hidden partitioning (automatic data organization for efficient querying).
- Time travel (querying past data versions for auditing).
- Atomic transactions (reliable concurrent operations).

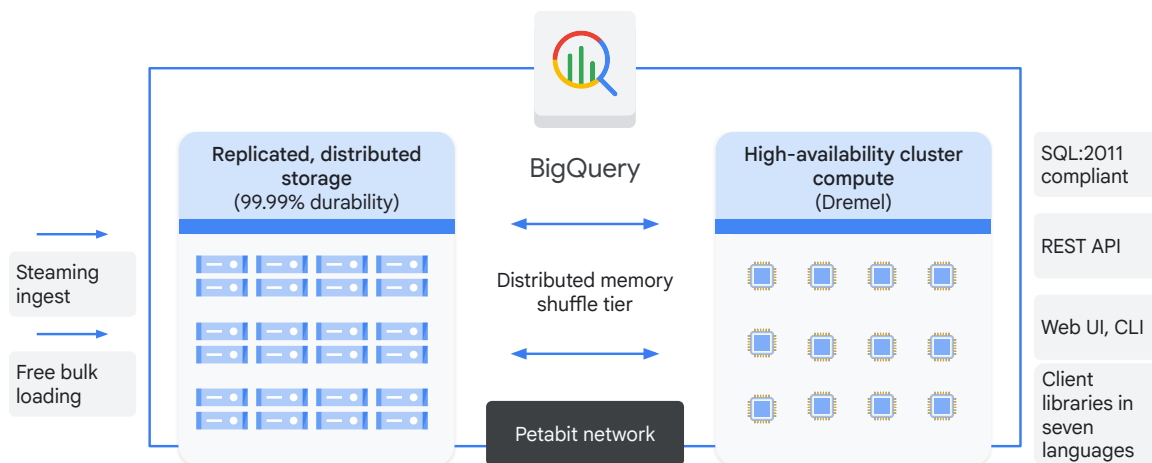
BigQuery

Enterprise-grade, fully managed,
serverless data warehouse.



BigQuery is Google's fully managed, serverless enterprise-grade data warehouse, serving as the central analytics and query engine.

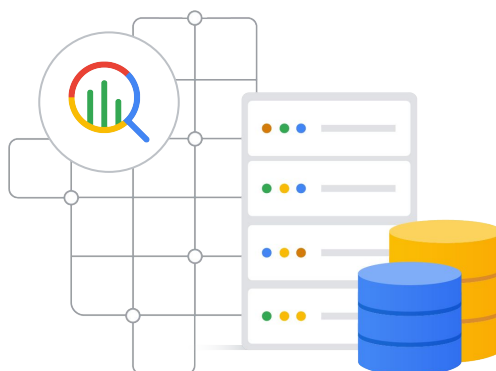
BigQuery fundamentals (1 of 2)



Its architecture separates storage from compute, enabling independent scaling and fast querying via the Dremel engine.

BigQuery fundamentals (2 of 2)

Through Lakehouse technology, BigQuery can directly query open-format data (such as Apache Iceberg tables) in Cloud Storage or other clouds, applying BigQuery's powerful optimizations, such as partitioning and clustering, by leveraging Apache Iceberg's metadata.



Through Lakehouse technology, BigQuery can directly query open-format data (such as Apache Iceberg tables) in Cloud Storage or other clouds, applying BigQuery's powerful optimizations, such as partitioning and clustering, by leveraging Iceberg's metadata.

Lakehouse

01

Allows you to create Lakehouse tables over data in object stores (such as Cloud Storage).

02

External tables reference data lake files, enabling unified governance and performance without duplicating data storage.

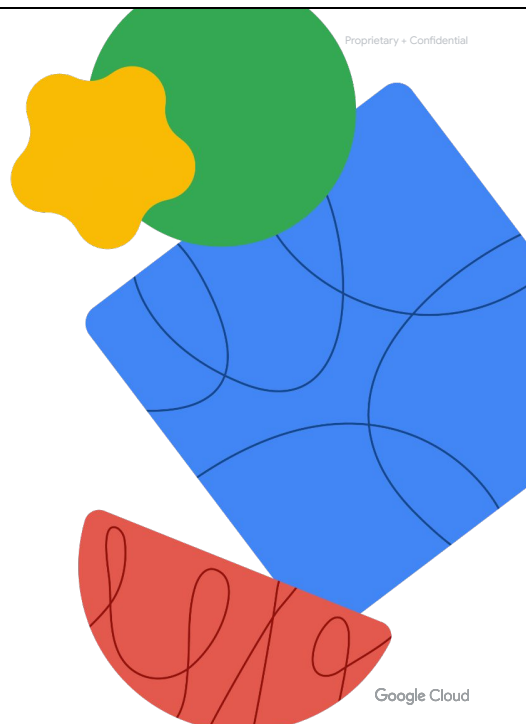


Lakehouse centralizes governance, extending BigQuery by enabling you to create Lakehouse tables over data in object stores, such as Cloud Storage.

These are external tables that don't store data within BigQuery; instead, they reference the source data files in your data lake, allowing you to query them with unified governance and performance.

Knowledge Catalog

Unified data fabric and centralized metadata hub for discovering, managing, monitoring, and governing data across the data lakehouse.

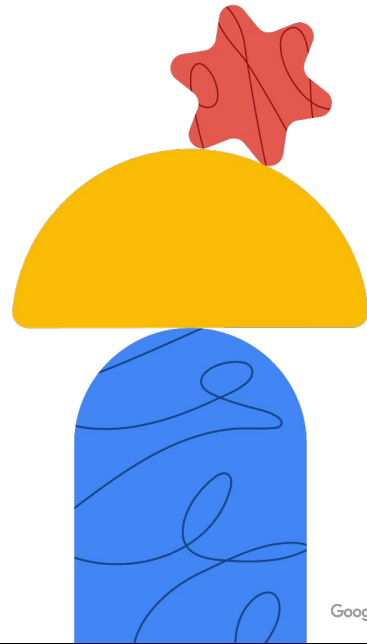


Google Cloud

You also learned that Knowledge Catalog is a unified data fabric and centralized metadata hub for discovering, managing, monitoring, and governing data across the entire data lakehouse, ensuring data quality and consistency.

Sensitive Data Protection

Automates the discovery, classification, and protection of sensitive data using techniques, such as masking, tokenization, or redaction.

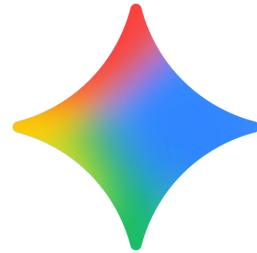


Google Cloud

Sensitive Data Protection, on the other hand, automates the discovery, classification, and protection of sensitive data (e.g., Personally Identifiable Information, such as email addresses, phone numbers, or credit card numbers) using techniques, such as masking, tokenization, or redaction.

BigQuery ML and Agent Platform

Enable advanced analytics and ML
directly on data lakehouse data.



BigQuery ML and Agent Platform enable advanced analytics and ML directly on data lakehouse data.

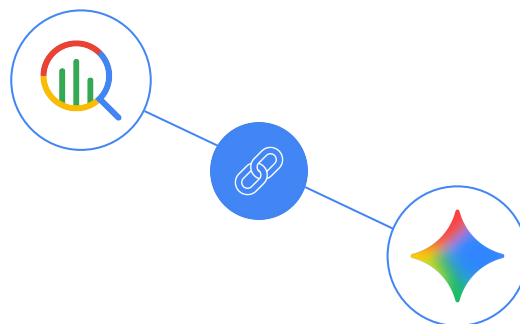
BigQuery ML and Agent Platform

01

BigQuery ML allows data analysts to build and deploy ML models using familiar SQL queries.

02

Agent Platform offers a comprehensive platform for complex custom models and MLOps, fully integrated with BigQuery.



BigQuery ML allows data analysts to build and deploy ML models using familiar SQL queries, whereas Agent Platform provides a comprehensive platform for more complex custom models and MLOps, seamlessly integrated with BigQuery data.



Course summary

In this course, you learned about...

- 01 The role and responsibilities of a data engineer.
- 02 Data engineering tasks and core components used on Google Cloud.
- 03 Creating and deploying data pipelines of varying patterns on Google Cloud.
- 04 Various automation techniques on Google Cloud to complete data engineering tasks.



Google Cloud

This concludes the course **Introduction to Data Engineering on Google Cloud**.

In this course, you learned about the duties and responsibilities of a data engineer.

How to identify data engineering tasks and core components used on Google Cloud to accomplish those tasks.

How to create and deploy data pipelines of varying patterns on Google Cloud.

And how to identify and utilize various automation techniques on Google Cloud to complete data engineering tasks.

We hope you had a great learning experience.

Data Engineering on Google Cloud



Introduction to Data Engineering on Google Cloud



Build Data Lakes and Data Warehouses on Google Cloud



Build Batch Data Pipelines on Google Cloud



Build Streaming Data Pipelines on Google Cloud



You've completed **Introduction to Data Engineering on Google Cloud**.

Build Data Lakes and Data Warehouses on Google Cloud is the second course of Data Engineering on Google Cloud.

Thank you for attending this training!

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